LU：

化列梯得U
L係數變號直接寫

$A=LU$
$U=L^{-1}A$

$(\quad)A=U$
$A=(\quad)^{-1}U$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ L_{21} & 1 & 0 & 0 \\ L_{31} & L_{32} & 1 & 0 \\ L_{41} & L_{42} & L_{43} & 1 \end{bmatrix} \cdot \begin{bmatrix} U_{11} & U_{12} & U_{13} & U_{14} \\ 0 & U_{22} & U_{23} & U_{24} \\ 0 & 0 & U_{33} & U_{34} \\ 0 & 0 & 0 & U_{44} \end{bmatrix}$$

$a_{11}=U_{11}$　　$a_{21}=L_{21}\cdot U_{11}$

$a_{12}=U_{12}$　　$a_{22}=L_{21}\cdot U_{12}+U_{22}$

$\Rightarrow a_{xy}=\sum_{i=1}^{n} L_{xi}U_{iy}$

Ex：

$$A=\begin{bmatrix} 2 & 4 & -1 & 5 \\ -4 & -5 & 3 & -8 \\ 2 & -5 & -4 & 1 \\ 4 & 2 & -4 & 7 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 & -1 & 5 \\ -4 & -5 & 3 & -8 \\ 2 & -5 & -4 & 1 \\ 4 & 2 & -4 & 7 \end{bmatrix} \xrightarrow{E_{r_{12}}^{2}E_{r_{13}}^{-1}E_{r_{14}}^{-2}} \begin{bmatrix} 2 & 4 & -1 & 5 \\ 0 & 3 & 1 & 2 \\ 0 & -9 & -3 & -4 \\ 0 & -6 & -2 & -3 \end{bmatrix} \xrightarrow{E_{r_{23}}^{3}E_{r_{24}}^{2}} \begin{bmatrix} 2 & 4 & -1 & 5 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{E_{r_{34}}^{-1/2}} \begin{bmatrix} 2 & 4 & -1 & 5 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

$$(E_{r_{12}}^{2}E_{r_{13}}^{-1}E_{r_{14}}^{-2}E_{r_{23}}^{3}E_{r_{24}}^{2}E_{r_{34}}^{-1/2})A=U$$

$$A=(E_{r_{12}}^{2}E_{r_{13}}^{-1}E_{r_{14}}^{-2}E_{r_{23}}^{3}E_{r_{24}}^{2}E_{r_{34}}^{-1/2})^{-1}U$$

$$L=(E_{r_{12}}^{2}E_{r_{13}}^{-1}E_{r_{14}}^{-2}E_{r_{23}}^{3}E_{r_{24}}^{2}E_{r_{34}}^{-1/2})^{-1}$$

$$=E_{r_{12}}^{-2}E_{r_{13}}^{1}E_{r_{14}}^{2}E_{r_{23}}^{-3}E_{r_{24}}^{-2}E_{r_{34}}^{1/2}$$

$$L=\begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ 2 & -2 & 1/2 & 1 \end{bmatrix}$$

Cholesky　給正定矩陣用的LU分解

$$A=L\cdot L^{T}=R^{T}\cdot R$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} L_{11} & 0 & 0 \\ L_{21} & L_{22} & 0 \\ L_{31} & L_{32} & L_{33} \end{bmatrix} \cdot \begin{bmatrix} L_{11} & L_{21} & L_{31} \\ 0 & L_{22} & L_{32} \\ 0 & 0 & L_{33} \end{bmatrix} = \begin{bmatrix} L_{11}^{2} & 0 & 0 \\ L_{21}\cdot L_{11} & L_{21}^{2}+L_{22}^{2} & 0 \\ L_{31}\cdot L_{11} & L_{31}\cdot L_{31}+L_{32}\cdot L_{32} & L_{31}^{2}+L_{32}^{2}+L_{33}^{2} \end{bmatrix}$$

以3x3為例

*Related：matlab chol()

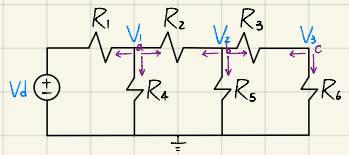
① 能快速檢驗正定矩陣的方法
⇒不是的話 報錯 or 出現複數
② 比LU分解的效率好

$$\begin{cases} L_{jj}=\dfrac{(a_{ij}-\sum_{k=1}^{j-1}L_{ik}L_{jk})}{L_{jj}}=\dfrac{(a_{ij}-\sum_{k=1}^{j-1}L_{ik}L_{jk})}{\sqrt{a_{jj}-(\sum_{k=1}^{j-1}L_{jk})^{2}}} & \forall i>j \\ L_{jj}=\sqrt{a_{jj}-(\sum_{k=1}^{j-1}L_{jk})^{2}} & \forall i=j \end{cases}$$

$$A=L\cdot D\cdot L^{T}=\begin{bmatrix} 1 & 0 & 0 \\ L_{21} & 1 & 0 \\ L_{31} & L_{32} & 1 \end{bmatrix}\cdot\begin{bmatrix} D_{11} & 0 & 0 \\ 0 & D_{22} & 0 \\ 0 & 0 & D_{33} \end{bmatrix}\cdot\begin{bmatrix} 1 & L_{21} & L_{31} \\ 0 & 1 & L_{32} \\ 0 & 0 & 1 \end{bmatrix}$$

Nodal Analysis (NA)

- KCL: node $I_{in} = I_{out}$
- KVL: closed loop $\Sigma V = 0$



$$\frac{V_1 - V_d}{R_1} + \frac{V_1 - V_2}{R_2} + \frac{V_1}{R_4} = 0$$

$$\frac{V_2 - V_1}{R_2} + \frac{V_2 - V_3}{R_3} + \frac{V_2}{R_5} = 0$$

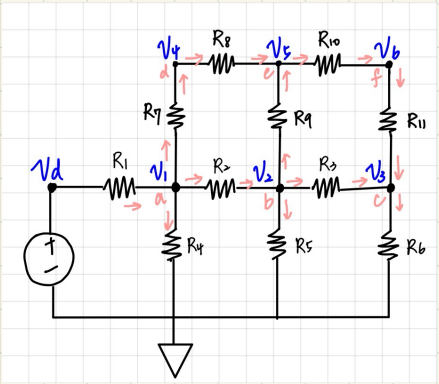
$$\frac{V_3 - V_2}{R_3} + \frac{V_3}{R_6} = 0$$

Source 移 = 右邊

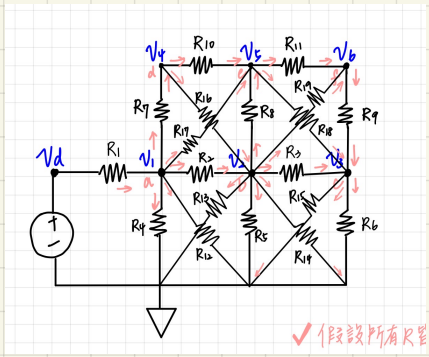
$$\begin{bmatrix} \frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_4} & -\frac{1}{R_2} & 0 \\ -\frac{1}{R_2} & \frac{1}{R_2}+\frac{1}{R_3}+\frac{1}{R_5} & -\frac{1}{R_3} \\ 0 & -\frac{1}{R_3} & \frac{1}{R_3}+\frac{1}{R_6} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} \frac{V_d}{R_1} \\ 0 \\ 0 \end{bmatrix}$$

和其它桌中間隔的R 倒數加負號

和本身桌相連的R 倒數相加



$$\begin{bmatrix} \frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_4}+\frac{1}{R_7} & -\frac{1}{R_2} & 0 & -\frac{1}{R_7} & 0 & 0 \\ -\frac{1}{R_2} & \frac{1}{R_2}+\frac{1}{R_3}+\frac{1}{R_5}+\frac{1}{R_9} & -\frac{1}{R_3} & 0 & -\frac{1}{R_9} & 0 \\ 0 & -\frac{1}{R_3} & \frac{1}{R_3}+\frac{1}{R_6}+\frac{1}{R_{11}} & 0 & 0 & -\frac{1}{R_{11}} \\ -\frac{1}{R_7} & 0 & 0 & \frac{1}{R_7}+\frac{1}{R_8} & -\frac{1}{R_8} & 0 \\ 0 & -\frac{1}{R_9} & 0 & -\frac{1}{R_8} & \frac{1}{R_8}+\frac{1}{R_9}+\frac{1}{R_{10}} & -\frac{1}{R_{10}} \\ 0 & 0 & -\frac{1}{R_{11}} & 0 & -\frac{1}{R_{10}} & \frac{1}{R_{10}}+\frac{1}{R_{11}} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \\ V_6 \end{bmatrix} = \begin{bmatrix} \frac{V_d}{R_1} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$



✓ 假設所有R皆

$$\begin{bmatrix} \frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_4}+\frac{1}{R_7}+\frac{1}{R_{12}}+\frac{1}{R_{17}} & -\frac{1}{R_2} & 0 & -\frac{1}{R_7} & -\frac{1}{R_{17}} & 0 \\ -\frac{1}{R_2} & \frac{1}{R_2}+\frac{1}{R_3}+\frac{1}{R_5}+\frac{1}{R_8}+\frac{1}{R_{13}}+\frac{1}{R_{15}}+\frac{1}{R_{16}}+\frac{1}{R_{19}} & -\frac{1}{R_3} & -\frac{1}{R_{16}} & -\frac{1}{R_8} & -\frac{1}{R_{19}} \\ 0 & -\frac{1}{R_3} & \frac{1}{R_3}+\frac{1}{R_6}+\frac{1}{R_9}+\frac{1}{R_{15}}+\frac{1}{R_{18}} & 0 & -\frac{1}{R_{18}} & -\frac{1}{R_9} \\ -\frac{1}{R_7} & -\frac{1}{R_{16}} & 0 & \frac{1}{R_7}+\frac{1}{R_{10}}+\frac{1}{R_{16}} & -\frac{1}{R_{10}} & 0 \\ -\frac{1}{R_{17}} & -\frac{1}{R_8} & -\frac{1}{R_{18}} & -\frac{1}{R_{10}} & \frac{1}{R_8}+\frac{1}{R_{10}}+\frac{1}{R_{11}}+\frac{1}{R_{16}}+\frac{1}{R_{19}} & -\frac{1}{R_{11}} \\ 0 & -\frac{1}{R_{19}} & -\frac{1}{R_9} & 0 & -\frac{1}{R_{11}} & \frac{1}{R_9}+\frac{1}{R_{11}}+\frac{1}{R_{19}} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \\ V_6 \end{bmatrix} = \begin{bmatrix} \frac{V_d}{R_1} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

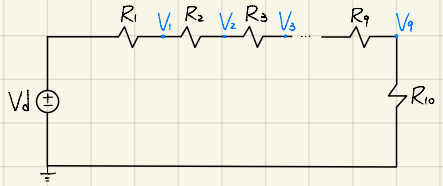
* 特性:

1. 對稱 (symmetry)
2. 對角線皆 > 0
3. eig(V) 皆 > 0, 為正定矩陣. (特徵值)
4. trace(A) > |其他 element 的 sum| (對角線 sum)

* Related: matlab diag( ), Sparse( )

$$x^T A x > 0$$

$$A = A^T$$



$$\begin{array}{c} \\ V_1 \\ V_2 \\ V_3 \\ V_4 \end{array} \begin{array}{c} \begin{array}{cccc} V_1 & V_2 & V_3 & V_4 \end{array} \\ \begin{bmatrix} \frac{1}{R_1}+\frac{1}{R_2} & -\frac{1}{R_2} & 0 & 0 \\ -\frac{1}{R_2} & \frac{1}{R_2}+\frac{1}{R_3} & -\frac{1}{R_3} & \\ 0 & -\frac{1}{R_3} & \frac{1}{R_3}+\frac{1}{R_4} & \\ 0 & & & \ddots \end{bmatrix} \end{array}$$

tridiagonal matrix

Laplace

$$F(s) = \int_{0^-}^{\infty} f(t) e^{-st} dt$$

$$f'(t) = SF(s) - f(0)$$

$$\int_0^t f(\tau) d\tau = \frac{1}{S} F(s)$$

$$(f * g)(t) = \int_0^t f(\tau) g(t-\tau) d\tau = F(s) \times G(s)$$

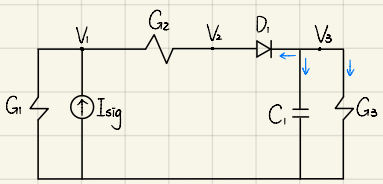
$$t \cdot u(t) = \frac{1}{S^2}$$

$$e^{-\alpha t} u(t) = \frac{1}{S+\alpha}$$

$$\delta(t) = 1$$

$$\delta(t-\tau) = e^{-\tau s}$$

補, MNA (Modified Nodal Analysis)



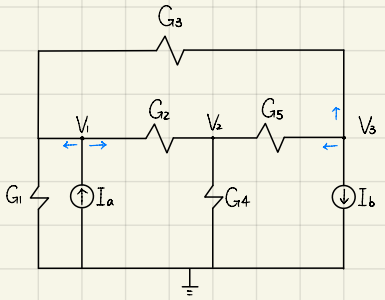
$$\begin{cases} G_1V_1 + G_2(V_1 - V_2) = I_{sig} \\ G_2(V_2 - V_1) + I_s(e^{\frac{V_2 - V_3}{V_t}} - 1) = 0 \\ G_3V_3 + C_1\dot{V}_3 - I_s(e^{\frac{V_2 - V_3}{V_t}} - 1) = 0 \end{cases}$$

$$\begin{aligned} (G_1 + G_2)V_1 - G_2V_2 &= I_{sig} \\ -G_2V_1 + G_2V_2 + I_s(e^{\frac{V_2 - V_3}{V_t}} - 1) &= 0 \\ G_3V_3 + C_1\dot{V}_3 - I_s(e^{\frac{V_2 - V_3}{V_t}} - 1) &= 0 \end{aligned}$$

$$\underbrace{\begin{bmatrix} G_1+G_2 & -G_2 & 0 \\ -G_2 & G_2 & 0 \\ 0 & 0 & G_3 \end{bmatrix}}_{\text{memoryless element}} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & C_1 \end{bmatrix}}_{\text{memory element}} \underbrace{\begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \\ \dot{V}_3 \end{bmatrix}}_{\dot{X}(t)} + \underbrace{\begin{bmatrix} 0 \\ I_s(e^{\frac{V_2 - V_3}{V_t}} - 1) \\ -I_s(e^{\frac{V_2 - V_3}{V_t}} - 1) \end{bmatrix}}_{\text{vector of nonlinear algebraic function}} = \underbrace{\begin{bmatrix} I_{sig} \\ 0 \\ 0 \end{bmatrix}}_{\text{vector of input source}}$$

$$G\underline{x(t)} + C\dot{x}(t) + f(x(t)) = b(t)$$

vector of unknowns

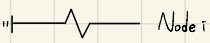


$$\begin{cases} G_1V_1 + G_2(V_1 - V_2) + G_3(V_1 - V_3) = I_a \\ G_2(V_2 - V_1) + G_4V_2 + G_5(V_2 - V_3) = 0 \\ G_3(V_3 - V_1) + G_5(V_3 - V_2) = -I_b \end{cases}$$

$$\begin{aligned} (G_1 + G_2 + G_3)V_1 - G_2V_2 - G_3V_3 &= I_a \\ -G_2V_1 + (G_2 + G_4 + G_5)V_2 - G_5V_3 &= 0 \\ -G_3V_1 - G_5V_2 + (G_3 + G_5)V_3 &= -I_b \end{aligned}$$

$$\begin{bmatrix} G_1+G_2+G_3 & -G_2 & -G_3 \\ -G_2 & G_2+G_4+G_5 & -G_5 \\ -G_3 & -G_5 & G_3+G_5 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} I_a \\ 0 \\ -I_b \end{bmatrix}$$

Resister Stamp - G Matrix



row i

col i

$$\begin{bmatrix} \ddots & \vdots & \\ \cdots & g & \cdots \\ & \vdots & \ddots \end{bmatrix}$$

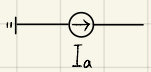
row i

row j

col i col j

$$\begin{bmatrix} \ddots & & \\ & g \cdots -g & \\ & \vdots \ddots \vdots & \\ & -g \cdots g & \\ & & \ddots \end{bmatrix}$$

Current Source Stamp, in RHS vector

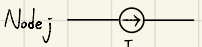


row i

$$\begin{bmatrix} \vdots \\ I_a \\ \vdots \end{bmatrix}$$

流出: -

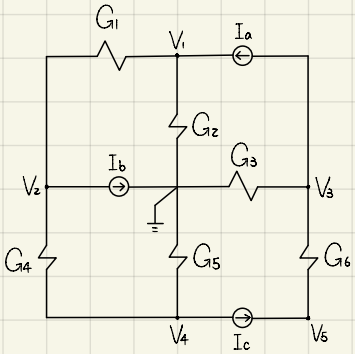
流入: +



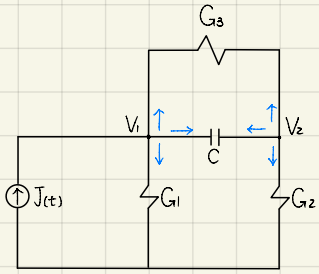
row i

row j

$$\begin{bmatrix} \vdots \\ I_a \\ \vdots \\ -I_a \\ \vdots \end{bmatrix}$$



$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} \begin{array}{c} \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{bmatrix} G_1+G_2 & -G_1 & 0 & 0 & 0 \\ -G_1 & G_1+G_4 & 0 & -G_4 & 0 \\ 0 & 0 & G_3+G_6 & 0 & -G_6 \\ 0 & -G_4 & 0 & G_4+G_5 & 0 \\ 0 & 0 & -G_6 & 0 & G_6 \end{bmatrix} \end{array} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \end{bmatrix} = \begin{bmatrix} I_a \\ -I_b \\ -I_a \\ -I_c \\ I_c \end{bmatrix}$$



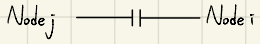
$$G_1V_1 + C(\dot{V}_1 - \dot{V}_2) + G_3(V_1 - V_2) = J(t)$$
$$G_2V_2 + C(\dot{V}_2 - \dot{V}_1) + G_3(V_2 - V_1) = 0$$

$$G \quad x(t) + C \quad \dot{x}(t) = b(t)$$

$$\begin{bmatrix} G_1+G_3 & -G_3 \\ -G_3 & G_3+G_2 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} + \begin{bmatrix} C & -C \\ -C & C \end{bmatrix}\begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix} = \begin{bmatrix} J(t) \\ 0 \end{bmatrix}$$

Capacitor Stamp - C Matrix

$$\begin{array}{c} \\ \text{row } i \\ \text{row } j \end{array}\begin{array}{c} \begin{array}{cc} \text{col } i & \text{col } j \end{array} \\ \begin{bmatrix} \ddots & \\ C & -C \\ -C & C \\ & \ddots \end{bmatrix} \end{array}$$



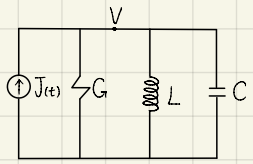
$$\begin{array}{c} \\ \text{row } i \end{array}\begin{array}{c} \begin{array}{c} \text{col } i \end{array} \\ \begin{bmatrix} \ddots & \\ C & \\ & \ddots \end{bmatrix} \end{array}$$

Resister Model : $I = GV$ $\quad V = IR$

$Q = CV = It$

Capacitor Model : $I(t) = C\frac{dv}{dt}$ $\quad V(t) = \frac{1}{C}\int_0^t I(t)d\tau + V(0)$

Inductor Model : $I(t) = \frac{1}{L}\int_0^t V(t)d\tau + V(0)$ $\quad V(t) = L\frac{dI}{dt}$

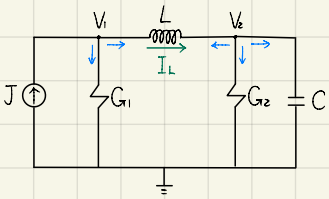


$$GV(t) + C\dot{V}(s) + \frac{1}{L}\int_0^t V(\tau)d\tau = J(t)$$

$$\xrightarrow{\mathcal{L}} \Rightarrow GV(s) + sCV(s) + \frac{1}{sL}V(s) = J(s)$$

if $s \to 0$, $\frac{1}{sL} \to \infty$

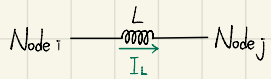
X work



$$G_1V_1 + I_L = J(t)$$
$$G_2V_2 + C\dot{V}_2 - I_L = 0$$
$$V_1 - V_2 = L\dot{I}_L$$

$$\begin{bmatrix} G_1 & 0 & 1 \\ 0 & G_2 & -1 \\ 1 & -1 & 0 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \\ I_L \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & C & 0 \\ 0 & 0 & -L \end{bmatrix}\begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \\ \dot{I}_L \end{bmatrix} = \begin{bmatrix} J(t) \\ 0 \\ 0 \end{bmatrix}$$

Inductor Stamp

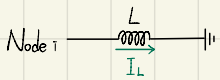


$$\begin{array}{c} \\ \text{row } i \\ \text{row } j \\ I_L \end{array}\begin{array}{c} \begin{array}{ccc} \text{col } i & \text{col } j & I_L \end{array} \\ \begin{bmatrix} \ddots & \cdots & \vdots \\ & & 1 \\ & & -1 \\ \cdots & 1 \cdots & -1 \quad 0 \end{bmatrix} \end{array}$$

G Matrix

$$\begin{array}{c} \\ \text{row } i \\ \text{row } j \\ I_L \end{array}\begin{array}{c} \begin{array}{ccc} \text{col } i & \text{col } j & I_L \end{array} \\ \begin{bmatrix} \ddots & \cdots & \vdots \\ & & 0 \\ & & 0 \\ \cdots & 0 \cdots & 0 \quad -L \end{bmatrix} \end{array}$$

C Matrix

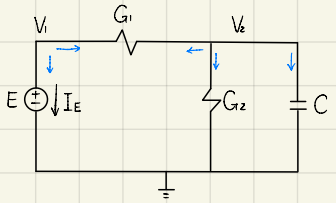


$$\begin{array}{c} \\ \text{row } i \\ \text{row } j \\ I_L \end{array}\begin{array}{c} \begin{array}{ccc} \text{col } i & \text{col } j & I_L \end{array} \\ \begin{bmatrix} \ddots & \cdots & \vdots \\ & & 1 \\ & & 0 \\ \cdots & 1 \cdots & 0 \quad 0 \end{bmatrix} \end{array}$$

G Matrix

$$\begin{array}{c} \\ \text{row } i \\ \text{row } j \\ I_L \end{array}\begin{array}{c} \begin{array}{ccc} \text{col } i & \text{col } j & I_L \end{array} \\ \begin{bmatrix} \ddots & \cdots & \vdots \\ & & 0 \\ & & 0 \\ \cdots & 0 \cdots & 0 \quad -L \end{bmatrix} \end{array}$$

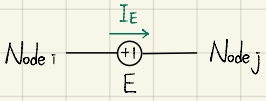
C Matrix

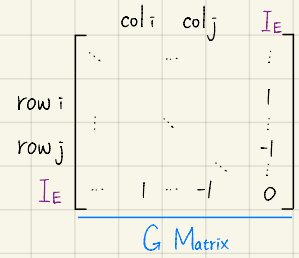


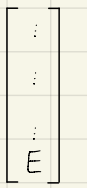
$$G_1(V_1 - V_2) + I_E = 0$$
$$G_1(V_2 - V_1) + G_2 V_2 + C\dot{V}_2 = 0$$
$$V_1 = E$$

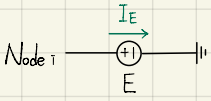
$$\begin{bmatrix} G_1 & -G_1 & 1 \\ -G_1 & G_1+G_2 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ I_E \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & C & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \\ \dot{I}_E \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ E \end{bmatrix}$$

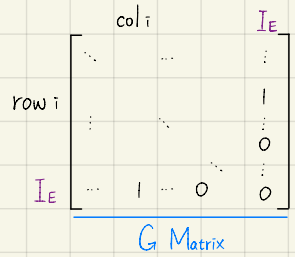
Voltage Source Stamp

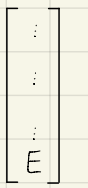




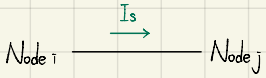


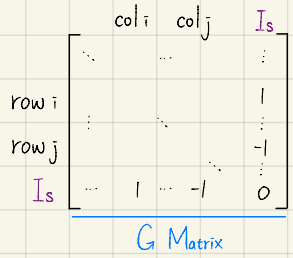






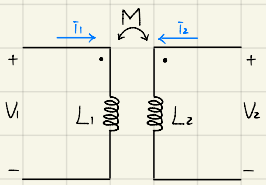
Short Circuit Stamp





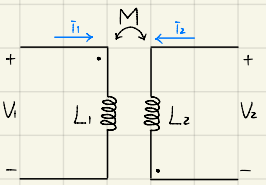
Mutual Inductane Model

互助



$$V_1 = L_1\frac{dI_1}{dt} + M\frac{dI_2}{dt}$$
$$V_2 = L_2\frac{dI_2}{dt} + M\frac{dI_1}{dt}$$
$$M = k\sqrt{L_1 L_2} \qquad 0 \le k \le 1$$
無耦合

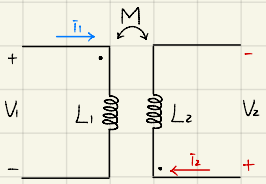
互消



$$V_1 = L_1\frac{dI_1}{dt} - M\frac{dI_2}{dt}$$
$$V_2 = L_2\frac{dI_2}{dt} - M\frac{dI_1}{dt}$$
$$M = k\sqrt{L_1 L_2} \qquad 0 \le k \le 1$$
無耦合

⇓

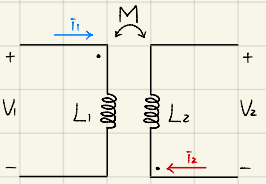
互消



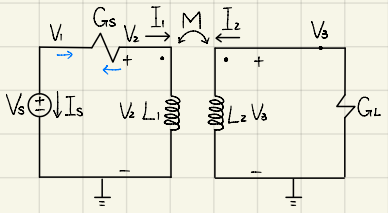
$$V_1 = L_1\frac{dI_1}{dt} + M\frac{dI_2}{dt}$$
$$V_2 = L_2\frac{dI_2}{dt} + M\frac{dI_1}{dt}$$
$$M = k\sqrt{L_1 L_2} \qquad 0 \le k \le 1$$
無耦合

⇐

互消



$$V_1 = L_1\frac{dI_1}{dt} + M\frac{dI_2}{dt}$$
$$V_2 = -L_2\frac{dI_2}{dt} - M\frac{dI_1}{dt}$$
$$M = k\sqrt{L_1 L_2} \qquad 0 \le k \le 1$$
無耦合

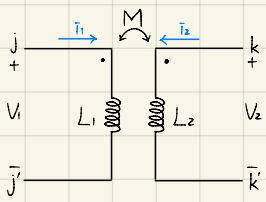


$$G_s(V_1 - V_2) + I_s = 0$$
$$G_s(V_2 - V_1) + I_1 = 0$$
$$G_L V_3 + I_2 = 0$$
$$V_1 = V_s$$

$$V_2 = L_1 \frac{dI_1}{dt} + M \frac{dI_2}{dt}$$
$$V_3 = L_2 \frac{dI_2}{dt} + M \frac{dI_1}{dt}$$

$$\begin{array}{c} \\ V_1 \\ V_2 \\ V_3 \\ I_s \\ I_1 \\ I_2 \end{array} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -L_1 & -M \\ 0 & 0 & 0 & 0 & -M & -L_1 \end{bmatrix} \cdot \begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \\ \dot{V}_3 \\ \dot{I}_s \\ \dot{I}_1 \\ \dot{I}_2 \end{bmatrix} + \begin{array}{c} \\ V_1 \\ V_2 \\ V_3 \\ I_s \\ I_1 \\ I_2 \end{array} \begin{array}{c} \begin{array}{cccccc} V_1 & V_2 & V_3 & & & \end{array} \\ \begin{bmatrix} G_s & -G_s & 0 & 1 & & \\ -G_s & G_s & 0 & 0 & 1 & \\ 0 & 0 & G_L & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ & 1 & 0 & 0 & 0 & 0 \\ & & 1 & 0 & 0 & 0 \end{bmatrix} \end{array} \cdot \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ i_L \\ i_{s1} \\ i_{s2} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ V_s \\ 0 \\ 0 \end{bmatrix}$$

Mutual Inductane Stamp C Matrix



$$\begin{array}{c} \\ \text{row } j \\ \text{row } j' \\ \text{row } k \\ \text{row } k' \\ \bar{I}_1 \\ \bar{I}_2 \end{array} \begin{array}{c} \begin{array}{cccccc} \text{col } j & \text{col } j' & \text{col } k & \text{col } k' & \bar{I}_1 & \bar{I}_2 \end{array} \\ \begin{bmatrix} \ddots & \cdots & \cdots & \cdots & \vdots & \\ \vdots & \ddots & & & 1 & \\ & & \ddots & & -1 & \\ & & & \ddots & \vdots & 1 \\ \cdots & 1 & \cdots & -1 & & \vdots \\ & \cdots & & 1 \cdots & -1 & -1 \\ & & & & & 0 \end{bmatrix} \end{array}$$

G Matrix

$$\begin{array}{c} \\ \text{row } j \\ \text{row } j' \\ \text{row } k \\ \text{row } k' \\ \bar{I}_1 \\ \bar{I}_2 \end{array} \begin{array}{c} \begin{array}{cccccc} \text{col } j & \text{col } j' & \text{col } k & \text{col } k' & \bar{I}_1 & \bar{I}_2 \end{array} \\ \begin{bmatrix} \ddots & \cdots & \cdots & \cdots & \vdots & \\ \vdots & \ddots & & & & \\ & & \ddots & & & \\ & & & \ddots & & \\ \cdots & & & & -L_1 & -M \\ & \cdots & & & -M & -L_2 \end{bmatrix} \end{array}$$

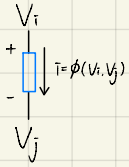
C Matrix

Nonliner R Stamp

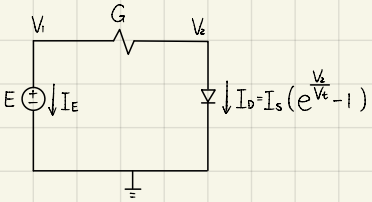
Linear

$$I_s\left(e^{\frac{V_i - V_j}{V_t}} - 1\right)$$

Nonlinear

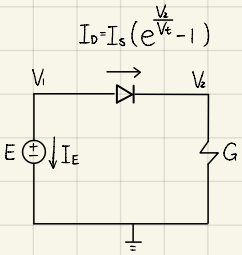


$$\begin{array}{c} \\ \text{row } i \\ \text{row } j \end{array} \begin{bmatrix} \vdots \\ \phi(V_i, V_j) \\ \vdots \\ -\phi(V_i, V_j) \\ \vdots \end{bmatrix}$$



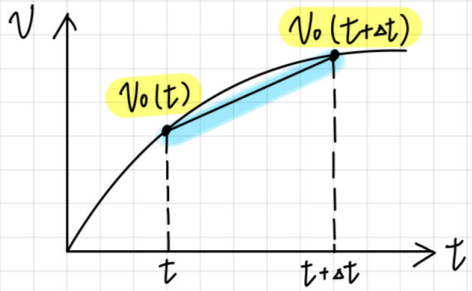
$$G(V_1 - V_2) + I_E = 0$$
$$G(V_2 - V_1) + I_s\left(e^{\frac{V_2}{V_t}} - 1\right) = 0$$
$$V_1 = E$$

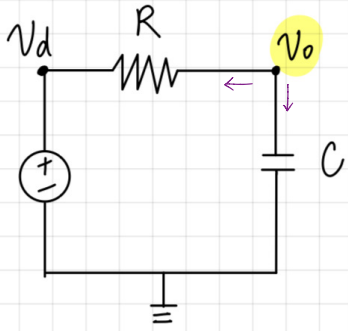
$$\begin{bmatrix} G & -G & 1 \\ -G & G & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ I_E \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ I_s\left(e^{\frac{V_2}{V_t}} - 1\right) \\ 0 \end{bmatrix}}_{\text{nonlinear vector}} = \begin{bmatrix} 0 \\ 0 \\ E \end{bmatrix}$$



$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & G & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ I_E \end{bmatrix} + \underbrace{\begin{bmatrix} I_s\left(e^{\frac{V_1 - V_2}{V_t}} - 1\right) \\ -I_s\left(e^{\frac{V_1 - V_2}{V_t}} - 1\right) \\ 0 \end{bmatrix}}_{\text{nonlinear vector}} = \begin{bmatrix} 0 \\ 0 \\ E \end{bmatrix}$$

Euler Method (一种一階数值方法，用以對常微分方程求解.)





$Q = CV = I_t$

$I = C \cdot \frac{dV}{dt}$

1. Forward Euler:

$$\frac{V_o - V_d}{R} + C \cdot \frac{dV_o}{dt} = 0$$

V_o 代入 $V_o(t)$ ⇒

$$C \cdot \frac{dV_o}{dt} = \frac{V_d - V_o}{R}$$

$$C \cdot \frac{V_o(t+\Delta t) - V_o(t)}{\Delta t} = \frac{V_d - V_o(t)}{R}$$

$$RC \cdot (V_o(t+\Delta t) - V_o(t)) = (V_d - V_o(t))\Delta t$$

$$V_o(t+\Delta t) = \frac{\Delta t}{RC}(V_d - V_o(t)) + V_o(t)$$

$$= \frac{\Delta t}{RC} V_d + \left(1 - \frac{\Delta t}{RC}\right) V_o(t)$$

Assume $V_o(0)=0$, $V_d=1$, $R=1$, $C=1$, then:

$$V_{(0.1)} = \frac{0.1}{1} \cdot 1 + \left(1 - \frac{0.1}{1}\right) \cdot 0 = 0.1$$

$$V_{(0.2)} = \frac{0.1}{1} \cdot 1 + \left(1 - \frac{0.1}{1}\right) \cdot 0.1 = 0.1 + 0.9 \times 0.1 = 0.19$$

$$V_{(0.3)} = \frac{0.1}{1} \cdot 1 + \left(1 - \frac{0.1}{1}\right) \cdot 0.19 = 0.1 + 0.9 \times 0.19 = 0.271$$

2. Backward Euler

$$\frac{V_o - V_d}{R} + C \cdot \frac{dV_o}{dt} = 0$$

V_o 代入 $V_o(t+\Delta t)$ ⇒

$$C \cdot \frac{dV_o}{dt} = \frac{V_d - V_o}{R}$$

$$C \cdot \frac{V_o(t+\Delta t) - V_o(t)}{\Delta t} = \frac{V_d - V_o(t+\Delta t)}{R}$$

$$RC \cdot (V_o(t+\Delta t) - V_o(t)) = (V_d - V_o(t+\Delta t))\Delta t$$

$$V_o(t+\Delta t)(RC + \Delta t) = V_d \cdot \Delta t + RC \cdot V_o(t)$$

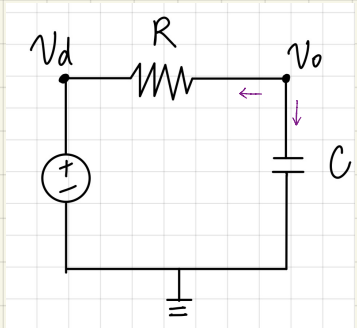
$$V_o(t+\Delta t) = \frac{\frac{\Delta t}{RC} \cdot V_d + V_o(t)}{1 + \frac{\Delta t}{RC}}$$

Assume $V_o(0)=0$, $V_d=1$, $R=1$, $C=1$, then:

$$V_{(0.1)} = \frac{\frac{0.1}{1} \cdot 1 + 0}{1 + \frac{0.1}{1}} = \frac{0.1}{1.1} = \frac{1}{11} \approx 0.09$$

$$V_{(0.2)} = \frac{\frac{0.1}{1} \cdot 1 + \frac{1}{11}}{1 + \frac{0.1}{1}} = \frac{21}{121} \approx 0.173$$

$$V_{(0.3)} = \frac{\frac{0.1}{1} \cdot 1 + \frac{21}{121}}{1 + \frac{0.1}{1}} = \frac{31}{1331} \approx 0.248$$



3. Laplace Transform

$$\frac{V_o - V_d}{R} + C \cdot \frac{dV_o}{dt} = 0$$

$$\mathcal{L}\left\{\frac{V_o - V_d}{R}\right\} + \mathcal{L}\left\{C \cdot \frac{dV_o}{dt}\right\} = 0$$

令 V_d 為 unit impulse func. $\delta(t)$

$$\frac{V_{o(s)} - V_{d(s)}}{R} + C \cdot S \cdot V_{o(s)} = 0$$

$$V_{o(s)} - V_{d(s)} + RC \cdot S \cdot V_{o(s)} = 0$$

$$V_{o(s)} + RC \cdot S \cdot V_{o(s)} = V_{d(s)}$$

$$V_{o(s)} = \frac{V_{d(s)}}{1 + SRC}$$

$$= \frac{\frac{V_{d(s)}}{RC}}{S + \frac{1}{RC}}$$

令 V_d 為 unit step func. $u(t)$

$$\frac{V_{o(s)} - \frac{V_{d(s)}}{S}}{R} + C \cdot S \cdot V_{o(s)} = 0$$

$$V_{o(s)} - \frac{V_{d(s)}}{S} + RC \cdot S \cdot V_{o(s)} = 0$$

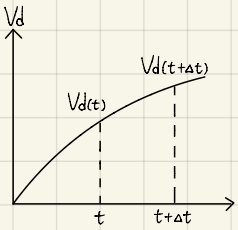
$$V_{o(s)} + RC \cdot S \cdot V_{o(s)} = \frac{V_{d(s)}}{S}$$

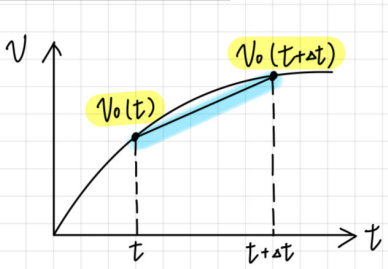
$$V_{o(s)} = \frac{\frac{V_{d(s)}}{S}}{1 + SRC}$$

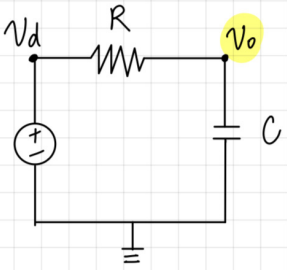
$$= \frac{\frac{V_{d(s)}}{SRC}}{S + \frac{1}{RC}}$$

令 $V_{d(s)} = 1$

$$= \frac{1}{S} + \frac{-1}{S + \frac{1}{RC}}$$







1. Forward Euler: ✓

$$\frac{V_o - V_d}{R} + C\cdot\frac{dV_o}{dt} = 0$$

V_o 代入 $V_o(t)$ ⇒ V_d 代入 $V_d(t)$

$$C\cdot\frac{dV_o}{dt} = \frac{V_d - V_o}{R}$$

$$C\cdot\frac{V_o(t+\Delta t) - V_o(t)}{\Delta t} = \frac{V_d(t) - V_o(t)}{R}$$

$$RC\cdot(V_o(t+\Delta t) - V_o(t)) = (V_d(t) - V_o(t))\Delta t$$

$$V_o(t+\Delta t) = \frac{\Delta t}{RC}(V_d(t) - V_o(t)) + V_o(t)$$

$$= \frac{\Delta t}{RC}V_d(t) + \left(1 - \frac{\Delta t}{RC}\right)V_o(t)$$

2. Backward Euler ✓

$$\frac{V_o - V_d}{R} + C\cdot\frac{dV_o}{dt} = 0$$

V_o 代入 $V_o(t+\Delta t)$ ⇒ V_d 代入 $V_d(t+\Delta t)$

$$C\cdot\frac{dV_o}{dt} = \frac{V_d - V_o}{R}$$

$$C\cdot\frac{V_o(t+\Delta t) - V_o(t)}{\Delta t} = \frac{V_d(t+\Delta t) - V_o(t+\Delta t)}{R}$$

$$RC\cdot(V_o(t+\Delta t) - V_o(t)) = (V_d(t+\Delta t) - V_o(t+\Delta t))\Delta t$$

$$V_o(t+\Delta t)(RC+\Delta t) = V_d\cdot\Delta t + RC\cdot V_o(t)$$

$$V_o(t+\Delta t) = \frac{\frac{\Delta t}{RC}\cdot V_d(t+\Delta t) + V_o(t)}{1 + \frac{\Delta t}{RC}}$$

Trapezoidal

$$\frac{V_o - V_d}{R} + C\cdot\frac{dV_o}{dt} = 0$$

$$C\cdot\frac{dV_o}{dt} = \frac{V_d - V_o}{R}$$

$$C\cdot\frac{V_o(t+\Delta t) - V_o(t)}{dt} = \frac{\frac{V_d(t)+V_d(t+\Delta t)}{2} - \frac{V_o(t)+V_o(t+\Delta t)}{2}}{R}$$

$$V_o(t+\Delta t) - V_o(t) = \frac{dt}{2RC}(V_d(t)+V_d(t+\Delta t)) - \frac{dt}{2RC}V_o(t) - \frac{dt}{2RC}V_o(t+\Delta t)$$

$$\left(1 + \frac{dt}{2RC}\right)V_o(t+\Delta t) = \frac{dt}{2RC}(V_d(t)+V_d(t+\Delta t)) + \left(1 - \frac{dt}{2RC}\right)V_o(t)$$

Program

$$C\frac{dV(t)}{dt} + GV(t) = I$$

BE:

$$C\cdot\frac{V_o(t) - V_o(t-1)}{\Delta t} + GV_o(t) = I$$

$$\left(\frac{C}{\Delta t} + G\right)V_o(t) = \frac{C}{\Delta t}V_o(t-1) + I$$

FE:

$$C\cdot\frac{V_o(t) - V_o(t-1)}{\Delta t} + GV_o(t-1) = I$$

$$\frac{V_o(t) - V_o(t-1)}{\Delta t} = C^{-1}(I - GV_o(t-1))$$

$$V_o(t) - V_o(t-1) = \Delta t C^{-1}I - \Delta t C^{-1}GV_o(t-1)$$

$$V_o(t) = (I + \Delta t C^{-1}G)V_o(t-1) + \Delta t C^{-1}I$$

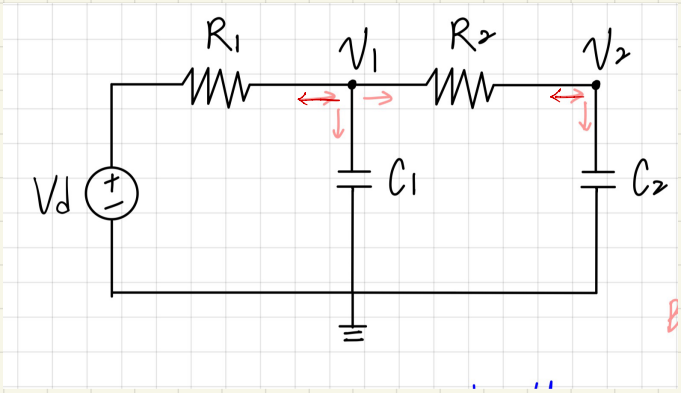
$$C\cdot\frac{V_o(t) - V_o(t-1)}{\Delta t} + GV_o(t) = I$$

$$\left(G + \frac{C}{\Delta t}\right)V_o(t) = \frac{C}{\Delta t}V_o(t-1) + I$$

$$C\cdot\frac{V_o(t) - V_o(t-1)}{\Delta t} + GV_o(t-1) = I$$

$$C(V_o(t) - V_o(t-1)) + \Delta t GV_o(t-1) = \Delta t I$$

$$V_o(t) = \frac{(C - \Delta t G)V_o(t-1) + \Delta t I}{C}$$



$$\begin{cases} \dfrac{V_1 - V_d}{R_1} + C_1 \cdot \dfrac{dV_1}{dt} + \dfrac{V_1 - V_2}{R_2} = 0 \\ C_2 \cdot \dfrac{dV_2}{dt} + \dfrac{V_2 - V_1}{R_2} = 0 \end{cases}$$

令 V_d 為 unit step func. $u(t)$

$$\xrightarrow{\mathcal{L}} \begin{cases} \dfrac{V_1(s) - V_d(s)}{R_1} + C_1 s V_1(s) + \dfrac{V_1(s) - V_2(s)}{R_2} = 0 \\ C_2 s V_2(s) + \dfrac{V_2(s) - V_1(s)}{R_2} = 0 \end{cases}$$

$$\begin{bmatrix} \frac{1}{R_1} + \frac{1}{R_2} + sC_1 & -\frac{1}{R_2} \\ -\frac{1}{R_2} & \frac{1}{R_2} + sC_2 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} \frac{V_d}{R_1} \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{R_1} + \frac{1}{R_2} & -\frac{1}{R_2} \\ -\frac{1}{R_2} & \frac{1}{R_2} \end{bmatrix} \begin{bmatrix} V_1(s) \\ V_2(s) \end{bmatrix} + s \begin{bmatrix} C_1 & 0 \\ 0 & C_2 \end{bmatrix} \begin{bmatrix} V_1(s) \\ V_2(s) \end{bmatrix} = \begin{bmatrix} \frac{V_d}{R_1} \\ 0 \end{bmatrix}$$

$$\left(\begin{bmatrix} \frac{1}{R_1} + \frac{1}{R_2} & -\frac{1}{R_2} \\ -\frac{1}{R_2} & \frac{1}{R_2} \end{bmatrix} + s \begin{bmatrix} C_1 & 0 \\ 0 & C_2 \end{bmatrix} \right) \begin{bmatrix} V_1(s) \\ V_2(s) \end{bmatrix} = \begin{bmatrix} \frac{V_d}{R_1} \\ 0 \end{bmatrix}$$

$$(G + sC)\, V_s(s) = I_s(s)$$

1. Eigen Value Method

$$(G + SC)V_s(s) = I_s(s)$$
$$G(I + SG^{-1}C)V_s(s) = I_s(s)$$

令 $A = G^{-1}C = W\Sigma W^{-1}$

$$G(I + SA)V_s(s) = I_s(s)$$
$$G(I + SW\Sigma W^{-1})V_s(s) = I_s(s)$$

eigen decomposition

$$(W \cdot W^{-1} + SW\Sigma W^{-1})V_s(s) = G^{-1}I_s(s)$$
$$W(I + S\Sigma)W^{-1}V_s(s) = G^{-1}I_s(s)$$
$$V_s(s) = W(I + S\Sigma)^{-1}W^{-1}G^{-1}I_s(s)$$
$$= W \begin{bmatrix} \frac{1}{1+S\lambda_1} & & \\ & \frac{1}{1+S\lambda_2} & \\ & & \ddots \end{bmatrix} W^{-1}G^{-1}I_s(s)$$

$Ax = \lambda x$

$W = [x_1\ x_2\ \cdots\ x_n]$ $\quad \Sigma = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$

$[Ax_1\ Ax_2 \cdots Ax_n] = [\lambda_1 x_1\ \lambda_2 x_2 \cdots \lambda_n x_n]$

$AW = W\Sigma$

$A = W\Sigma W^{-1}$

$A \cdot A^{-1} = I$

2. Moment matching

$$(G+SC)V_s(s) = I_s(s)$$
$$G(I+SG^{-1}C)V_s(s) = I_s(s)$$
令 $A = G^{-1}C$
$$G(I+SA)V_s(s) = I_s(s)$$
$$V_s(s) = (I+SA)^{-1}G^{-1}I_s(s)$$
$$= (I - SA + S^2A^2 - S^3A^3 + \cdots)G^{-1}I_s(s)$$

*Note: Taylor equation

$$\frac{1}{1+\delta} = 1 - \delta + \delta^2 - \delta^3 + \cdots$$
$$\frac{1}{1+\alpha s} = 1 - \alpha s + \alpha^2 s^2 - \alpha^3 s^3 + \cdots$$
$$\frac{Vd}{s+\frac{1}{\alpha}} \Rightarrow \alpha = RC$$

Let
$$m_0 = G^{-1}I_s(s) \quad \text{voltage}$$
$$m_1 = -G^{-1}Cm_0$$
$$m_2 = -G^{-1}Cm_1$$
$$\vdots$$
$$m_k = -G^{-1}Cm_k$$

$$V_s(s) = m_0 + m_1 s + m_2 s^2 + \cdots + m_k s^k$$

$$Gm_0 = I_s(s)$$
$$Gm_1 = -Cm_0$$
$$Gm_2 = -Cm_1$$

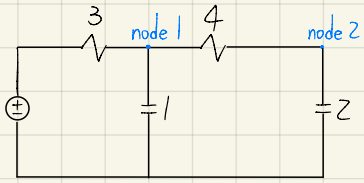
Suppose moment = 8

$$V_s(s) = (I+SA)^{-1}G^{-1}I_s(s) = m_0 + m_1 s + m_2 s^2 + \cdots + m_7 s^7$$
$$= \frac{b_0 + b_1 s + b_2 s^2 + \cdots + b_p s^p}{1 + a_1 s + a_2 s^2 + a_3 s^3 + \cdots + a_q s^q} \quad \begin{cases} p = q-1 \\ p+q = 7 \end{cases}$$
$$= \frac{b_0 + b_1 s + b_2 s^2 + b_3 s^3}{1 + a_1 s + a_2 s^2 + a_3 s^3 + a_4 s^4}$$

$$(1 + a_1 s + a_2 s^2 + a_3 s^3 + a_4 s^4)(m_0 + m_1 s + m_2 s^2 + \cdots + m_7 s^7) = b_0 + b_1 s + b_2 s^2 + b_3 s^3$$

	S	S^2	S^3	S^4	S^5	S^6	S^7	S^8	S^9	S^{10}	S^{11}
m_0 +	m_1 +	m_2 +	m_3 +	m_4 +	m_5 +	m_6 +	m_7				
	a_1m_0 +	a_1m_1 +	a_1m_2 +	a_1m_3 +	a_1m_4 +	a_1m_5 +	a_1m_6 +	a_1m_7			
		a_2m_0 +	a_2m_1 +	a_2m_2 +	a_2m_3 +	a_2m_4 +	a_2m_5 +	a_2m_6 +	a_2m_7		
			a_3m_0 +	a_3m_1 +	a_3m_2 +	a_3m_3 +	a_3m_4 +	a_3m_5 +	a_3m_6 +	a_3m_7	
				a_4m_0 +	a_4m_1 +	a_4m_2 +	a_4m_3 +	a_4m_4 +	a_4m_5 +	a_4m_6 +	a_4m_7
$\|\|$	$\|\|$	$\|\|$	$\|\|$								
b_0	b_1	b_2	b_3								

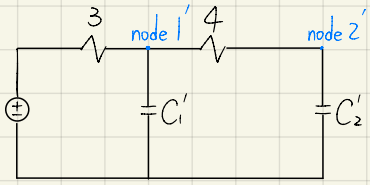
$$\begin{array}{c} \\ b_0 \\ b_1 \\ b_2 \\ b_3 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \end{array}\begin{array}{c} \begin{array}{cccccccc} b_0 & b_1 & b_2 & b_3 & a_1 & a_2 & a_3 & a_4 \end{array} \\ \begin{bmatrix} 1 & & & & & & & \\ & 1 & & & -m_0 & & & \\ & & 1 & & -m_1 & -m_0 & & \\ & & & 1 & -m_2 & -m_1 & -m_0 & \\ & & & & -m_3 & -m_2 & -m_1 & -m_0 \\ & & & & -m_4 & -m_3 & -m_2 & -m_1 \\ & & & & -m_5 & -m_4 & -m_3 & -m_2 \\ & & & & -m_6 & -m_5 & -m_4 & -m_3 \end{bmatrix} \end{array} \cdot \begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \begin{bmatrix} m_0 \\ m_1 \\ m_2 \\ m_3 \\ m_4 \\ m_5 \\ m_6 \\ m_7 \end{bmatrix}$$



Elmore Delay

node 1 = 3×(1+2) = 9

node 2 = 3×(1+2)+4×2 = 17



$C_1' = C_1 \times$ node 1 $= 1 \times 9 = 9$

$C_2' = C_2 \times$ node 2 $= 2 \times 17 = 34$

node 1' = 3×(C1'+C2') = 129

node 2' = 3×(C1'+C2')+4×C2' = 265

$1 - 9S + 129S^2 + \cdots$

$1 - 17S + 265S^2 + \cdots$

做 pade (m,n)

CG (Conjugate gradient method)

$$R=\rho\frac{l}{A}=\rho\frac{l}{Wt}$$

寬 ↓　↓ 厚

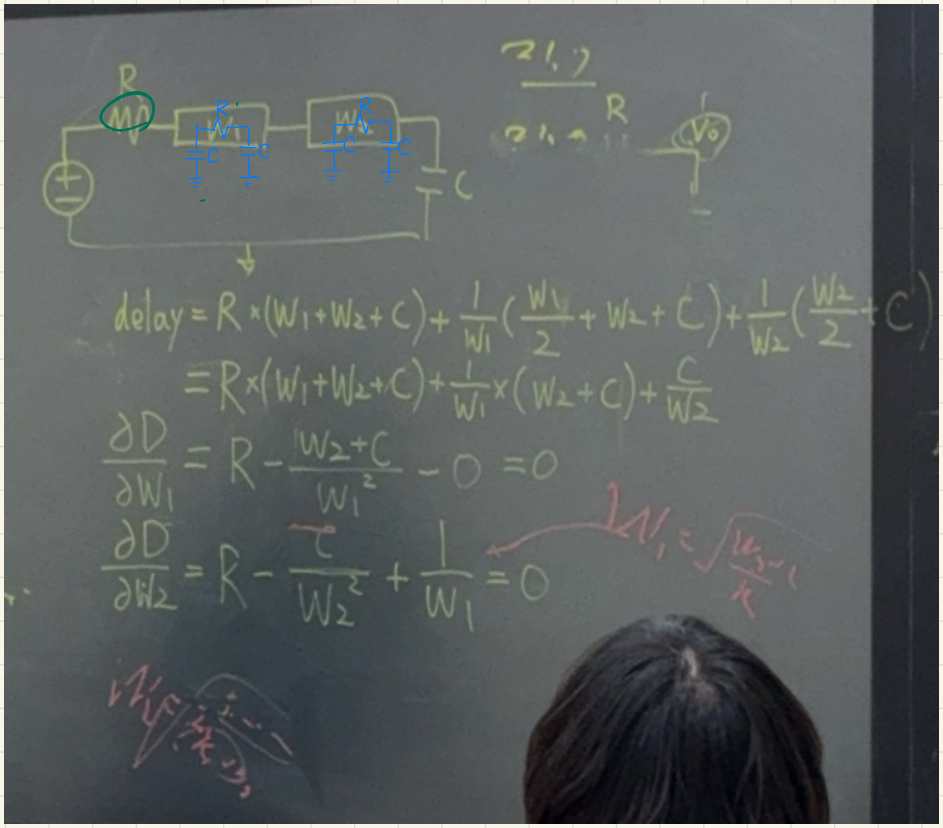
$$R\propto\frac{L}{W\cdot H}$$

$$C=\varepsilon\frac{A}{d}$$

A → wire area

d → wire to wire 夾出來

$$C=C'\times L\times W$$



$$\text{delay}(D)=R\times(W_1+W_2+C)+\frac{1}{W_1}\times\left(\frac{W_1}{2}+W_2+C\right)+\frac{1}{W_2}\times\left(\frac{W_2}{2}+C\right)$$
$$\approx R\times(W_1+W_2+C)+\frac{1}{W_1}\times(W_2+C)+\frac{C}{W_2}$$

找使 delay 最小的 W_1、W_2

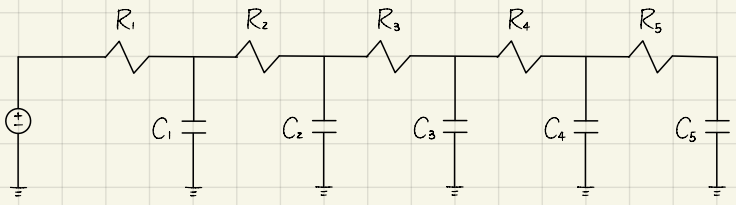
$$\frac{\partial D}{\partial W_1}=R-\frac{W_2+C}{W_1^2}=0 \qquad W_1^2=\frac{W_2+C}{R}$$

$$\frac{\partial D}{\partial W_2}=R+\frac{1}{W_1}-\frac{C}{W_2^2}=0 \qquad W_2=RW_1^2-C$$

$$R+\frac{1}{W_1}-\frac{C}{(RW_1^2-C)^2}=0$$

$$\left(R+\frac{1}{W_1}\right)(RW_1^2-C)^2-C=0$$

$$\left(R+\frac{1}{W_1}\right)(RW_1^2-C)(RW_1^2-C)-C=0$$



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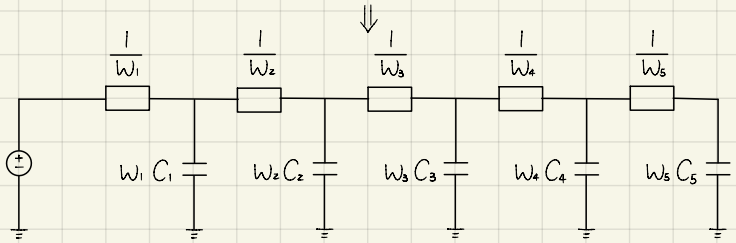
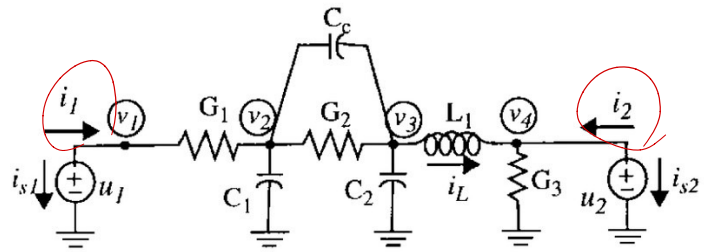


$$f(\vec{w})=\frac{1}{W_1}(W_1+W_2+W_3+W_4+W_5)+\frac{1}{W_2}(W_2+W_3+W_4+W_5)+\frac{1}{W_3}(W_3+W_4+W_5)$$
$$+\frac{1}{W_4}(W_4+W_5)+\frac{1}{W_5}W_5$$

R往後看所有C



$$
\begin{array}{c|cccc:ccc}
 & V_1 & V_2 & V_3 & V_4 & i_L & i_{s1} & i_{s2} \\
\hline
V_1 & 0 & 0 & 0 & 0 & & & \\
V_2 & 0 & (C_c+C_1) & -C_c & 0 & & & \\
V_3 & 0 & -C_c & (C_2+C_c) & 0 & & \mathbf{0} & \\
V_4 & 0 & 0 & 0 & 0 & & & \\
\hdashline
 & & & & & L_1 & 0 & 0 \\
 & & \mathbf{0} & & & 0 & 0 & 0 \\
 & & & & & 0 & 0 & 0
\end{array}
\begin{bmatrix} \dot{v}_1 \\ \dot{v}_2 \\ \dot{v}_3 \\ \dot{v}_4 \\ \dot{i}_L \\ \dot{i}_{s1} \\ \dot{i}_{s2} \end{bmatrix}
= -
\begin{array}{c|cccc:ccc}
 & & & & & & & \\
 & G_1 & -G_1 & 0 & 0 & 0 & 1 & 0 \\
 & -G_1 & (G_1+G_2) & -G_2 & 0 & 0 & 0 & 0 \\
 & 0 & -G_2 & G_2 & 0 & 1 & 0 & 0 \\
 & 0 & 0 & 0 & G_3 & -1 & 0 & 1 \\
\hdashline
 & 0 & 0 & -1 & 1 & & & \\
 & -1 & 0 & 0 & 0 & & \mathbf{0} & \\
 & 0 & 0 & 0 & -1 & & &
\end{array}
\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ i_L \\ i_{s1} \\ i_{s2} \end{bmatrix}
+
\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}
\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}
$$

$\mathbf{C}$ $\mathbf{G}$ $\mathbf{x}_n$ $\mathbf{B}$

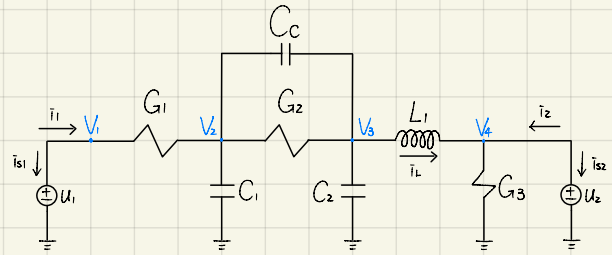
Port Currents

$$
\begin{bmatrix} i_1 \\ i_2 \end{bmatrix}
=
\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix}
\mathbf{x}_n
$$

$\mathbf{L^T}$

Port Voltages

頁



$$I = C\frac{dV}{dt} \qquad V = L\frac{dI}{dt}$$

$$(G+SC)V = I_s \qquad (G+SC)X_n = B\cdot u$$

$$C=\begin{bmatrix} Q & 0 \\ 0 & H \end{bmatrix} \quad G=\begin{bmatrix} N & E \\ -E^T & 0 \end{bmatrix} \quad x_n=\begin{bmatrix} V \\ i \end{bmatrix}$$

diagonal : node C 相加

off-diagonal : -(node C 相加)

diagonal : node R 相加

off-diagonal : -(node R 相加)

$i_L = V_3 - V_4$ 流出 : +

$i_{s1} = V_1 - 0$ 流入 : −

$i_{s2} = V_2 - 0$

$$
\begin{array}{c|cccc:ccc}
 & V_1 & V_2 & V_3 & V_4 & i_L & i_{s1} & i_{s2} \\
\hline
V_1 & 0 & 0 & 0 & 0 & & & \\
V_2 & 0 & (C_c+C_1) & -C_c & 0 & & & \\
V_3 & 0 & -C_c & (C_c+C_2) & 0 & & \mathbf{0} & \\
V_4 & 0 & 0 & 0 & 0 & & & \\
\hdashline
i_L & & & & & L_1 & 0 & 0 \\
i_{s1} & & \mathbf{0} & & & 0 & 0 & 0 \\
i_{s2} & & & & & 0 & 0 & 0
\end{array}
\begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \\ \dot{V}_3 \\ \dot{V}_4 \\ \dot{i}_L \\ \dot{i}_{s1} \\ \dot{i}_{s2} \end{bmatrix}
= -
\begin{array}{c|cccc:ccc:c}
 & V_1 & V_2 & V_3 & V_4 & i_L & i_{s1} & i_{s2} & E \\
\hline
V_1 & G_1 & -G_1 & 0 & 0 & & & & 1 \\
V_2 & -G_1 & (G_1+G_2) & -G_2 & 0 & & & & \\
V_3 & 0 & -G_2 & G_2 & 0 & 1 & & & \\
V_4 & 0 & 0 & 0 & G_3 & -1 & & & 1 \\
\hdashline
i_L & & & & & -1 & 1 & 0 & 0 \\
i_{s1} & -1 & & & & 0 & 0 & 0 & 0 \\
i_{s2} & & & & -1 & 0 & 0 & 0 & 0
\end{array}
\begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ i_L \\ i_{s1} \\ i_{s2} \end{bmatrix}
+
\begin{array}{c|cc}
 & V_1 & V_2 \\
\hline
 & 0 & 0 \\
 & 0 & 0 \\
 & 0 & 0 \\
 & 0 & 0 \\
 & 0 & 0 \\
i_{s1} & -1 & 0 \\
i_{s2} & 0 & -1
\end{array}
\cdot
\begin{bmatrix} U_1 \\ U_2 \end{bmatrix}
$$

$-E^T$

$V_1 = U_1$ $\quad V_1 - U_1 = 0$

$V_2 = U_2$ $\quad V_2 - U_2 = 0$

$$
\begin{bmatrix} I_1 \\ I_2 \end{bmatrix}
=
\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix}
X_n
$$

$I_1 = -i_{s1}$

$I_2 = -i_{s2}$

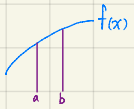
1. Given (G+sC)V = Is, Derive the solution for form the following method
   (a) Trapezoidal Method: V(t+Δt) =?
   (b) Eigen Decomposition: V(t) = ?
   (c) Moment Matching Method: V = ?

$$(G+SC)V=I_s$$

$$C\frac{dV(t)}{dt}+GV(t)=I_s$$

a. Trapezoidal Method:

$$\int_t^{t+\Delta t} C\frac{dV}{dt}dt+\int_t^{t+\Delta t} GV(t)dt=\int_t^{t+\Delta t} I_s dt$$



$$\int_a^b f(x)dx=\frac{(f(a)+f(b))\times(b-a)}{2}$$

$$C(V(t+\Delta t)-V(t))+G\frac{\Delta t(V(t+\Delta t)+V(t))}{2}=\frac{\Delta t(I_s(t+\Delta t)+I_s(t))}{2}$$

$$(C+G\frac{\Delta t}{2})V(t+\Delta t)=(C-G\frac{\Delta t}{2})V(t)+\frac{\Delta t}{2}(I_s(t+\Delta t)+I_s(t))$$

$$V(t+\Delta t)=(C+G\frac{\Delta t}{2})^{-1}\{(C-G\frac{\Delta t}{2})V(t)+\frac{\Delta t}{2}(I_s(t+\Delta t)+I_s(t))\}$$

b. Eigen Decomposition

$$(G+SC)V(S)=I_s(S)$$
$$G(I+SG^{-1}C)V(S)=I_s(S)$$

令 $A=G^{-1}C=W\Sigma W^{-1}$

$$G(I+SA)V(S)=I_s(S)$$
$$G(I+SW\Sigma W^{-1})V(S)=I_s(S)$$ Eigen decomposition
$$(WW^{-1}+SW\Sigma W^{-1})V(S)=G^{-1}I_s(S)$$
$$W(I+S\Sigma)W^{-1}V(S)=G^{-1}I_s(S)$$
$$V(S)=W(I+S\Sigma)^{-1}W^{-1}G^{-1}I_s(S)$$
$$=W\begin{bmatrix}\frac{1}{1+S\lambda_1} & & \\ & \frac{1}{1+S\lambda_2} & \\ & & \ddots\end{bmatrix}W^{-1}G^{-1}I_s(S)$$

$$\mathcal{L}^{-1}\{V(S)\}=\mathcal{L}^{-1}\left\{W\begin{bmatrix}\frac{1}{1+S\lambda_1} & & \\ & \frac{1}{1+S\lambda_2} & \\ & & \ddots\end{bmatrix}W^{-1}G^{-1}I_s(S)\right\}$$

$$\frac{\frac{1}{\lambda}}{S+\frac{1}{\lambda}}=\frac{1}{\lambda}e^{-\frac{t}{\lambda}}$$

$$e^{-at}=\mathcal{L}^{-1}\{\frac{1}{s+a}\}$$

$$V(t)=\int_0^t W\begin{bmatrix}\frac{1}{\lambda_1}e^{-\frac{t-\tau}{\lambda_1}} & & & \\ & \frac{1}{\lambda_2}e^{-\frac{t-\tau}{\lambda_2}} & & \\ & & \ddots & \\ & & & \frac{1}{\lambda_n}e^{-\frac{t-\tau}{\lambda_n}}\end{bmatrix}W^{-1}G^{-1}I_s(\tau)d\tau$$

$$\mathcal{L}\{h(t)*u(t)\}=H(s)U(s)$$

$$h(t)*u(t)=\int_0^t h(t-\tau)u(\tau)d\tau$$

C. moment matching

$$(G+SC)V_s(S)=I_s(S)$$
$$G(I+SG^{-1}C)V_s(S)=I_s(S)$$

令 $A=G^{-1}C$

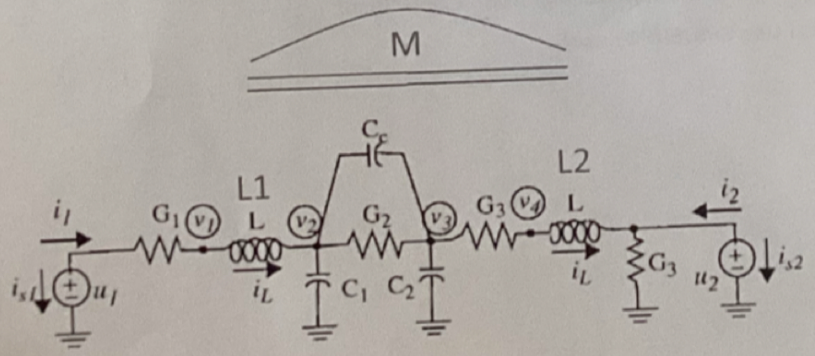
$$G(I+SA)V_s(S)=I_s(S)$$
$$V_s(S)=(I+SA)^{-1}G^{-1}I_s(S)$$
$$=(I-SA+S^2A^2-S^3A^3+\cdots)G^{-1}I_s(S)$$

Let $m_0=G^{-1}I_s(S)$
$m_1=-G^{-1}Cm_0$
$m_2=-G^{-1}Cm_1$
$\vdots$

$$V_s(S)=m_0+m_1S+m_2S^2+\cdots$$

$M = k\sqrt{L_1 L_2}$

2. Please derive the below circuit using following algorithm (k = $0.6\sqrt{L1L2}$)



(a) Use eigenvector method (with Inductance)

(b) Use eigenvector method (without Inductance)

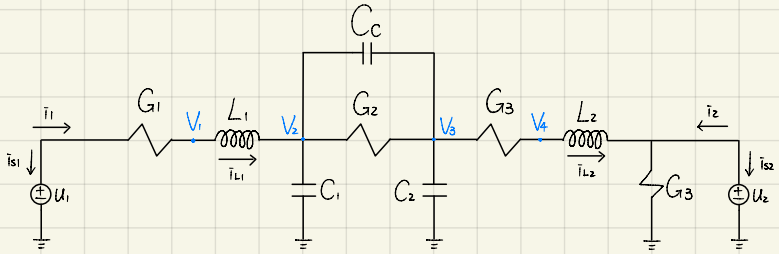
(c) Use trapezoidal method (with Inductance)

(d) Use trapezoidal method (without Inductance)

(e) Use moment matching method (without Inductance)

(f) Use moment matching method (with Inductance)

$(V_1 - U_1)G_1 + I_{L1} = 0$



$\bar{I}_{s1} + G_1(U_1 - V_1) = 0$

$-\frac{1}{G_1}\bar{I}_{s1} - U_1 + V_1 = 0$

$M = k\sqrt{L_1 L_2} = 0.6 L_1 L_2$

$\bar{I}_{L2} = U_2 G_3 + \bar{I}_{s2}$

$V_4 - L_2\frac{d\bar{I}_{L2}}{dt} - M\frac{d\bar{I}_{L1}}{dt} = U_2$

$$\begin{array}{c} \\ V_1 \\ V_2 \\ V_3 \\ V_4 \\ \dot{I}_{L1} \\ \dot{I}_{L2} \\ \dot{I}_{s1} \\ \dot{I}_{s2} \end{array}\begin{array}{c} \begin{array}{cccccccc} V_1 & V_2 & V_3 & V_4 & i_{L1} & i_{L2} & i_{s1} & i_{s2} \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 & & & & \\ 0 & (C_c+C_1) & -C_c & 0 & & & & \\ 0 & -C_c & (C_c+C_2) & 0 & & & & \\ 0 & 0 & 0 & 0 & & & & \\ & & & & L_1 & 0.6L_1L_2 & 0 & 0 \\ & & & & 0.6L_1L_2 & L_2 & 0 & 0 \\ & & & & 0 & 0 & 0 & 0 \\ & & & & 0 & 0 & 0 & 0 \end{bmatrix} \end{array} \cdot \begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \\ \dot{V}_3 \\ \dot{V}_4 \\ \dot{I}_{L1} \\ \dot{I}_{L2} \\ \dot{I}_{s1} \\ \dot{I}_{s2} \end{bmatrix} = C \quad \dot{X}_n =$$

$$- \begin{array}{c} \\ V_1 \\ V_2 \\ V_3 \\ V_4 \\ i_{L1} \\ i_{L2} \\ i_{s1} \\ i_{s2} \end{array}\begin{array}{c} \begin{array}{cccccccc} V_1 & V_2 & V_3 & V_4 & i_{L1} & i_{L2} & i_{s1} & i_{s2} \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & & 1 & \\ 0 & G_2 & -G_2 & 0 & -1 & & & \\ 0 & -G_2 & (G_2+G_3) & -G_3 & & & & \\ 0 & 0 & -G_3 & G_3 & & 1 & & \\ -1 & 1 & & & 0 & 0 & 0 & 0 \\ & & & -1 & 0 & 0 & 0 & 0 \\ -1 & & & & 0 & 0 & \frac{1}{G_1} & 0 \\ & & & & 0 & 0 & 0 & 0 \end{bmatrix} \end{array} \cdot \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ i_{L1} \\ i_{L2} \\ i_{s1} \\ i_{s2} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & -1 \\ -1 & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} U_1 \\ U_2 \end{bmatrix}$$

$$-G \qquad X_n + BU_n$$

$$\begin{bmatrix} \bar{I}_1 \\ \bar{I}_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{bmatrix} X_n$$

$$
\begin{array}{c} \\ V_1 \\ V_2 \\ V_3 \\ V_4 \\ i_{L1} \\ i_{L2} \end{array}
\begin{array}{c}
\begin{array}{cccccc} V_1 & V_2 & V_3 & V_4 & i_{L1} & i_{L2} \end{array} \\
\begin{bmatrix}
0 & 0 & 0 & 0 & & \\
0 & (C_c+C_1) & -C_c & 0 & & \\
0 & -C_c & (C_c+C_3) & 0 & & \\
0 & 0 & 0 & 0 & & \\
 & & & & L_1 & 0 \\
 & & & & 0 & L_2
\end{bmatrix}
\end{array}
\cdot
\begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \\ \dot{V}_3 \\ \dot{V}_4 \\ \dot{i}_{L1} \\ \dot{i}_{L2} \end{bmatrix}
=
-
\begin{array}{c} \\ V_1 \\ V_2 \\ V_3 \\ V_4 \\ i_{L1} \\ i_{L2} \end{array}
\begin{array}{c}
\begin{array}{cccccc} V_1 & V_2 & V_3 & V_4 & i_{L1} & i_{L2} \end{array} \\
\begin{bmatrix}
G_1 & 0 & 0 & 0 & 1 & \\
0 & G_2 & -G_2 & 0 & -1 & \\
0 & -G_2 & (G_2+G_3) & -G_3 & & \\
0 & 0 & -G_3 & G_3 & & 1 \\
-1 & 1 & & & 0 & 0 \\
 & & & -1 & 0 & 0
\end{bmatrix}
\end{array}
\cdot
\begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ i_{L1} \\ i_{L2} \end{bmatrix}
+
\begin{bmatrix} G_1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & -1 \end{bmatrix}
\cdot
\begin{bmatrix} U_1 \\ U_2 \end{bmatrix}
$$

$$
C \quad \dot{X}_n = \qquad -G \qquad X_n + B U_n
$$

$$
\begin{bmatrix} i_1 \\ i_2 \end{bmatrix}
=
\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -1 & 0
\end{bmatrix}
X_n
$$

$$
\bar{i}_{L2} = U_2 G_3 + \bar{i}_{S2}
$$

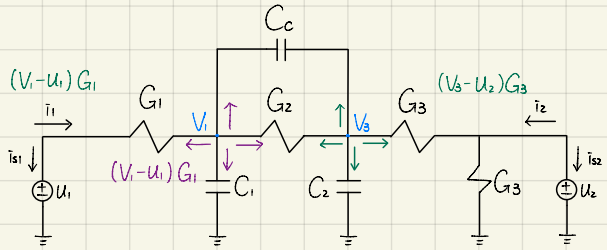
$$
\begin{array}{c} \\ V_1 \\ V_2 \\ V_3 \\ V_4 \\ i_{L1} \\ i_{L2} \end{array}
\begin{array}{c}
\begin{array}{cccccc} V_1 & V_2 & V_3 & V_4 & i_{L1} & i_{L2} \end{array} \\
\begin{bmatrix}
0 & 0 & 0 & 0 & & \\
0 & (C_c+C_1) & -C_c & 0 & & \\
0 & -C_c & (C_c+C_3) & 0 & & \\
0 & 0 & 0 & 0 & & \\
 & & & & L_1 & 0 \\
 & & & & 0 & L_2
\end{bmatrix}
\end{array}
\cdot
\begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \\ \dot{V}_3 \\ \dot{V}_4 \\ \dot{i}_{L1} \\ \dot{i}_{L2} \end{bmatrix}
=
-
\begin{array}{c} \\ V_1 \\ V_2 \\ V_3 \\ V_4 \\ i_{L1} \\ i_{L2} \end{array}
\begin{array}{c}
\begin{array}{cccccc} V_1 & V_2 & V_3 & V_4 & i_{L1} & i_{L2} \end{array} \\
\begin{bmatrix}
G_1 & 0 & 0 & 0 & 1 & \\
0 & G_2 & -G_2 & 0 & -1 & \\
0 & -G_2 & (G_2+G_3) & -G_3 & & \\
0 & 0 & -G_3 & G_3 & & 1 \\
-1 & 1 & & & 0 & 0 \\
 & & & -1 & 0 & 0
\end{bmatrix}
\end{array}
\cdot
\begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ i_{L1} \\ i_{L2} \end{bmatrix}
+
\begin{bmatrix} G_1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & -1 \end{bmatrix}
\cdot
\begin{bmatrix} U_1 \\ U_2 \end{bmatrix}
$$

$$
C \quad \dot{X}_n = \qquad -G \qquad X_n + B U_n
$$

$$
\bar{i}_{L2} = U_2 G_3 + \bar{i}_{S2}
$$

$$
\begin{bmatrix} \bar{i}_{S1} \\ \bar{i}_{S2} \\ \bar{i}_1 \\ \bar{i}_2 \end{bmatrix}
=
\begin{bmatrix}
0 & 0 & 0 & 0 & -1 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & -1
\end{bmatrix}
X_n +
\begin{bmatrix} 0 & 0 \\ 0 & -G_3 \\ 0 & 0 \\ 0 & G_3 \end{bmatrix}
\cdot
\begin{bmatrix} U_1 \\ U_2 \end{bmatrix}
$$

$$
\bar{i}_{L2} = G_3 U_2 + \bar{i}_{S2}
$$



$(V_1-U_1)G_1 = I_{S1}$

$(V_3-U_2)G_3 = I_{S2} + G_3U_2$

$$(G+SC)V_S(S) = I_S(S)$$

$$\begin{matrix} & V_1 & V_3 \end{matrix}$$

$$\left(\begin{matrix} V_1 \\ V_3 \end{matrix} \begin{bmatrix} G_1+G_2 & -G_2 \\ -G_2 & G_2+G_3 \end{bmatrix} + S \begin{matrix} V_1 \\ V_3 \end{matrix} \begin{bmatrix} C_1+C_c & -C_c \\ -C_c & C_2+C_c \end{bmatrix} \right) \begin{bmatrix} V_1 \\ V_3 \end{bmatrix} = \begin{bmatrix} U_1G_1 \\ U_2G_3 \end{bmatrix}$$

$$\begin{bmatrix} i_{S1} \\ i_{S2} \end{bmatrix} = \begin{bmatrix} G_1 & 0 \\ 0 & G_3 \end{bmatrix} \begin{bmatrix} V_1 \\ V_3 \end{bmatrix} + \begin{bmatrix} -G_1 & 0 \\ 0 & -2G_3 \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \end{bmatrix}$$

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} i_{S1} \\ i_{S2} \end{bmatrix}$$

(a)(b) eigenvector method

$$(G+SC)X_n = BU_N$$

$$G(I+SG^{-1}C)X_n = BU_N$$

令 $A = G^{-1}C = W\Sigma W^{-1}$

$$G(I+SA)X_n = BU_N$$

$$G(I+SW\Sigma W^{-1})X_n = BU_N$$ ← eigen decomposition

$$(WW^{-1}+SW\Sigma W^{-1})X_n = G^{-1}BU_N$$

$$W(I+S\Sigma)W^{-1}X_n = G^{-1}BU_N$$

$$X_n = W(I+S\Sigma)^{-1}W^{-1}G^{-1}BU_N$$

$$= W \begin{bmatrix} \frac{1}{1+S\lambda_1} & & \\ & \frac{1}{1+S\lambda_2} & \\ & & \ddots \end{bmatrix} W^{-1}G^{-1}BU_N$$ ，其中 W 為 $G^{-1}C$ eigenvector，Σ 為 $G^{-1}C$ eigenvalue

(e)(f) 令 $X_n = m_0 + m_1S + m_2S^2 + \cdots$

$$(G+SC)X_n = BU_n \qquad X_n = (G+SC)^{-1}BU_n$$

$$(G+SC)(m_0+m_1S+m_2S^2+\cdots) = BU_n$$

$$Gm_0 + (Gm_1+Cm_0)S + (Gm_2+Cm_1)S^2 \cdots = BU_n$$

$$Gm_0 = BU_n \qquad m_0 = G^{-1}BU_n$$

$$Gm_1 + Cm_0 = 0 \qquad m_1 = -G^{-1}Cm_0$$

$$Gm_2 + Cm_1 = 0 \qquad m_2 = -G^{-1}Cm_1$$

$$\vdots$$

$$X_n = m_0 + m_1S + m_2S^2 + \cdots = \frac{N(S)}{D(S)} = \frac{a_0+a_1S+a_2S^2}{1+b_1S+b_2S^2+b_3S^3}$$

1				$-m_0(\bar{1})$
1			$-m_1(\bar{1})$	$-m_0(\bar{1})$
1		$-m_2(\bar{1})$	$-m_1(\bar{1})$	$-m_0(\bar{1})$
	$-m_3(\bar{1})$	$-m_2(\bar{1})$	$-m_1(\bar{1})$	$-m_0(\bar{1})$
	$-m_4(\bar{1})$	$-m_3(\bar{1})$	$-m_2(\bar{1})$	$-m_1(\bar{1})$
	$-m_5(\bar{1})$	$-m_4(\bar{1})$	$-m_3(\bar{1})$	$-m_2(\bar{1})$
	$-m_6(\bar{1})$	$-m_5(\bar{1})$	$-m_4(\bar{1})$	$-m_3(\bar{1})$

$$m_0 = a_0$$

$$m_1 = a_1 - m_0b_1$$

$$m_2 = a_2 - m_1b_1 - m_0b_2$$

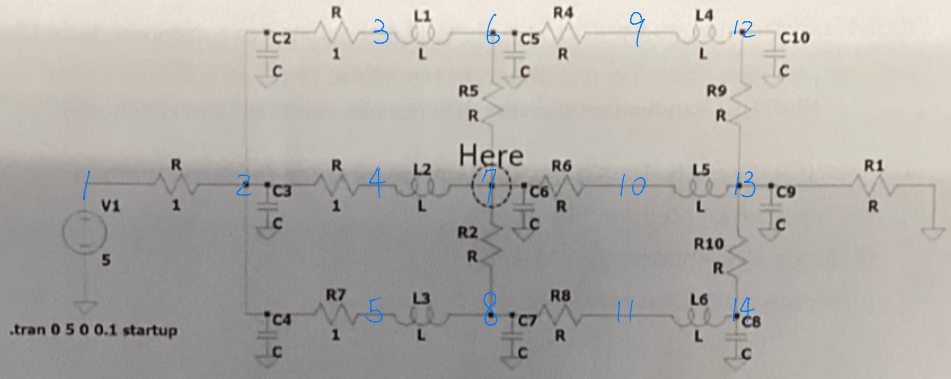
$$m_3 = a_3 - m_2b_1 - m_1b_2 - m_0b_3$$

$$m_4 = -m_3b_1 - m_2b_2 - m_1b_3 - m_0b_4$$

$$m_5 = -m_4b_1 - m_3b_2 - m_2b_3 - m_1b_4$$

[PART B] Programming

3



(a) Use backward Euler method (P3a.m)

(b) Use moment matching(order = 3) (P3b.m)

(c) Use SPICE (P3c.asc)

$$(G+SC)\,x(t) = Bu(t)$$

	V_1	V_2	V_3	V_4	V_5	V_6	V_7	V_8	i_{L1}	i_{L2}	i_{L3}	i_{L4}	i_{L5}	i_{L6}	i_{S1}		
V_1	1															=	V_1
V_2	-1	1														=	V_2
V_3																=	V_3
V_4																=	V_4
V_5																=	V_5
V_6																=	V_6
V_7																=	V_7
V_8																=	V_8
V_9																=	V_9
V_{10}																=	V_{10}
V_{11}																=	V_{11}
V_{12}																=	V_{12}
V_{13}																=	V_{13}
V_{14}																=	V_{14}
i_{L1}																=	i_{L1}
i_{L2}																=	i_{L2}
i_{L3}																=	i_{L3}
i_{L4}																=	i_{L4}
i_{L5}																=	i_{L5}
i_{L6}																=	i_{L6}
i_{S1}																=	i_{S1}

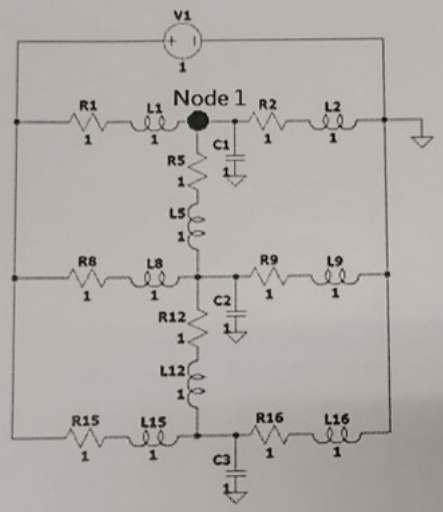
1. Reference the following sample 2 by 2 circuit with the following modifications.
Derive MNA for 100 by 100 circuit, all the voltage sources and resistance value are 1. Besides, there is a capacitance on each node (Node1~Node9999) and the capacitance value is 1, the inductors value is 1 as below(fig.1). Use the following methods to perform simulation at Node 1. Plot the result. You should find out voltages from t = 0 ~ t = 10 with time step = 0.1.

(a) (20%)Using trapezoid euler method to find Node 1. (P1_trap.m)

(b) (20%)Using moment matching to find Node 1. (P1_moment.m)

(c) (20%)Using LTspice to load the circuit(.sp file) and export Node 1. Plot it with (a)**You should write a code (in any programming language) to generate .sp file**

***Files to commit: P1_trap.m, P1_moment.m & P1.sp (Should contain a txt file from LTspice)**



(Note. 2 by 2 → 100 by 100)

fig.1

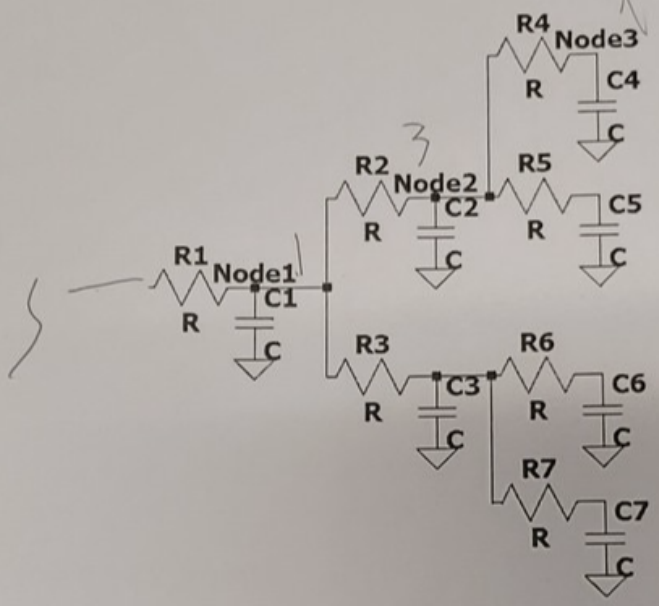


fig.3

We use L shape to model a wire segment (i.e. 1 resistor and 1 capacitance). Now given a network as fig.3 shows, there are 7 wire segments. The length of the segments are 1, all C and R are 1.

(a) (15%) Please write down the delay of Node1, Node2 and Node3 using Elmore delay model. Note that you should use W, L, R and C to represent the answer.

(b) (15%) Find out the W value(between 1 and 5) of all wire segments to minimize the delay of all leaf nodes (e.g. Node3). You should use the Matlab to find the answer. Please show **the width of each segment** and **the value of the best delay**.

1. Find Vx for the following circuit (fig1) by using the corresponding methods, plot the three results on the same figure on matlab.

   All the voltage, inductance, capacitance and resistance value is 1.

   To make the three plot consistent, you should find out Vx from t = 0 ~ t = 10 with time step = 0.1.

   (a) (10%) Using eigen method.

   (b) (10%) Using trapezoid euler method.

   (c) (15%)Using moment matching with all inductance replace with a short line.

   (d) (5%) Using LTspice to generate the circuit and export the voltage data.

   ***Files to commit: P1.m & P1.asc (Should contain a txt file from LTspice with any name.)**

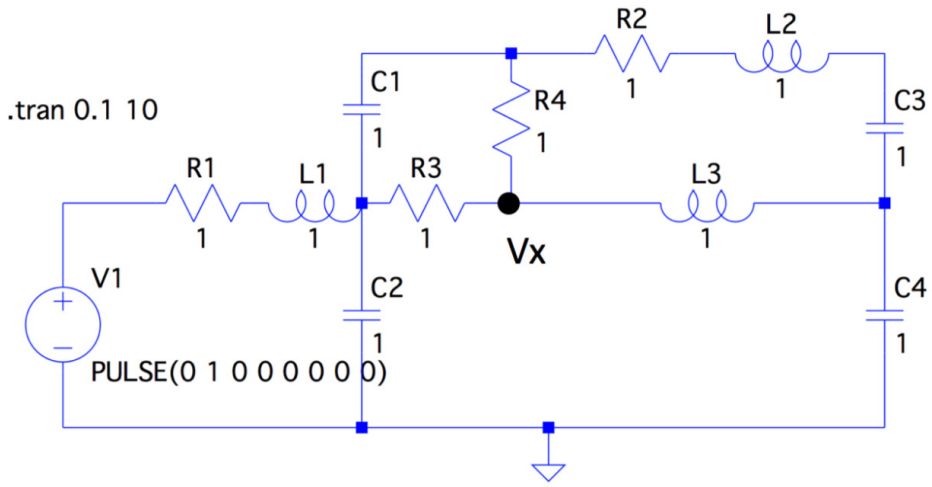


fig.1

1. MNA with Reactive Elements: Applying MNA to circuits with inductors ($v(t) = L \cdot \frac{di(t)}{dt}$) and capacitors ($i(t) = C \cdot \frac{dv(t)}{dt}$) presents no special difficulty if we use the complex impedance of these elements.

$$Z_R = R, Z_L = sL = j\omega L, Z_C = \frac{1}{sC} = -\frac{j}{\omega C}$$

That is, for equation $(G + sC)V = I$, we also need to modify the matrices. You can find a modified example in here: goo.gl/vBtgbT [1].

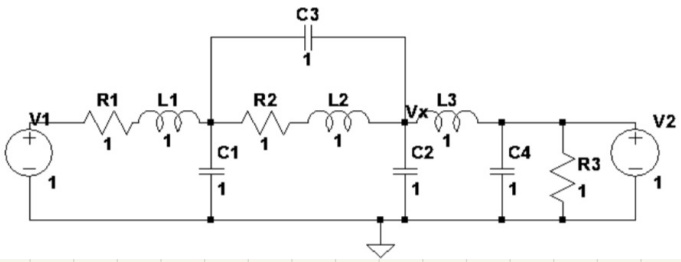
(a) (15%) Derive the MNA equations (matrix G, C, V and I) for the circuit below. Perform transient simulation at **Vx** by using **eigenvector method**. Let time step be 0.1s and simulate for 20s. Plot the results in one figure.

Replace all inductors (L1, L2 and L3) with resisters (R4, R5 and R6) and the values of them are **2**. Perform transient simulation at **Vx** by using **different methods below**. Let time step be 0.1s and simulate for 20s. Plot the results in one figure.

(b) (10%) Use **forward Euler method**.

(c) (10%) Use **trapezoidal method**.

(d) (10%) Use **SPICE**.



3. (15%) Derive LU Decomposition Prove that Gaussian elimination and LU Decomposition method are equivalent.

$$A = LU$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots \\ a_{21} & a_{22} & a_{23} & \cdots \\ a_{31} & a_{32} & a_{33} & \\ \vdots & & & \ddots \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \cdots \\ L_{21} & 1 & 0 & \cdots \\ L_{31} & L_{32} & 1 & \\ \vdots & & & \ddots \end{bmatrix} \cdot \begin{bmatrix} U_{11} & U_{12} & U_{13} & \cdots \\ 0 & U_{22} & U_{23} & \cdots \\ 0 & 0 & U_{33} & \\ \vdots & & & \ddots \end{bmatrix}$$

$$= \begin{bmatrix} U_{11} & U_{12} & U_{13} & \\ L_{21}\cdot U_{11} & L_{21}\cdot U_{12}+U_{22} & & \cdots \\ L_{31}\cdot U_{11} & L_{31}\cdot U_{12}+L_{32}\cdot U_{22} & U_{33} & \\ & & \vdots & \ddots \end{bmatrix} \quad \begin{matrix} 0 & 0 \\ 1 & 0 \\ L_{32} & 1 \end{matrix}$$

$$a_{11}=U_{11} \qquad a_{21}=L_{21}\cdot U_{11}$$
$$a_{12}=U_{12} \qquad a_{22}=L_{21}\cdot U_{12}+U_{22}$$
$$\Rightarrow a_{xy}=\sum_{i=1}^{n} L_{xi}U_{iy}$$

在高斯消去法中，使用消去倍數 $m_{ij}=\frac{a_{ij}}{a_{jj}}\ (i>j)$ 將对角線下方的元素消為0，
在LU分解中，L為紀錄高斯消去法的過程，U為上三角矩陣
（高斯消去矩陣乘積的反矩陣）

3. (a) (10%) Derive the **Cholesky decomposition** formulae without square root ($A = LDL^{-1}$), write down the general form of L and D. (The meaning of L and D, please refer to **Appendix 1**)

(b) (15%) Implement this algorithm as a Matlab function.

(Note that, Name the function as "LDLT", which takes an input and generates output pair [L U], I will test it with some cases.)

Files to commit: LDLT.m

$$A = L\cdot D\cdot L^{T} = \begin{bmatrix} 1 & 0 & 0 \\ L_{21} & 1 & 0 \\ L_{31} & L_{32} & 1 \end{bmatrix} \cdot \begin{bmatrix} D_{11} & 0 & 0 \\ 0 & D_{22} & 0 \\ 0 & 0 & D_{33} \end{bmatrix} \cdot \begin{bmatrix} 1 & L_{21} & L_{31} \\ 0 & 1 & L_{32} \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} D_{11} & 0 & 0 \\ L_{21}\cdot D_{11} & D_{22} & 0 \\ L_{31}\cdot D_{11} & L_{32}\cdot D_{22} & D_{33} \end{bmatrix} \cdot \begin{bmatrix} 1 & L_{21} & L_{31} \\ 0 & 1 & L_{32} \\ 0 & 0 & 1 \end{bmatrix}$$

$$A_{jj} = D_{jj} + \sum_{k=1}^{j-1} L_{jk}^{2} D_{kk}$$
$$\Rightarrow D_{jj} = A_{jj} - \sum_{k=1}^{j-1} L_{jk}^{2} D_{kk}$$

$$= \begin{bmatrix} D_{11} & & \\ L_{21}\cdot D_{11} & L_{21}^{2}\cdot D_{11}+D_{22} & \\ L_{31}\cdot D_{11} & L_{21}\cdot L_{31}\cdot D_{11}+L_{32}\cdot D_{22} & L_{31}^{2}\cdot D_{11}+L_{32}^{2}\cdot D_{22}+D_{33} \end{bmatrix}$$

(i: row, j: column; Symmetric)

$$A_{ij} = L_{ij}\cdot D_{jj} + \sum_{k=1}^{j-1} L_{ik} L_{jk} D_{kk}$$

$$A = L \cdot L^T = \begin{bmatrix} L_{11} & 0 & 0 \\ L_{21} & L_{22} & 0 \\ L_{31} & L_{32} & L_{33} \end{bmatrix} \cdot \begin{bmatrix} L_{11} & L_{21} & L_{31} \\ 0 & L_{22} & L_{32} \\ 0 & 0 & L_{33} \end{bmatrix}$$

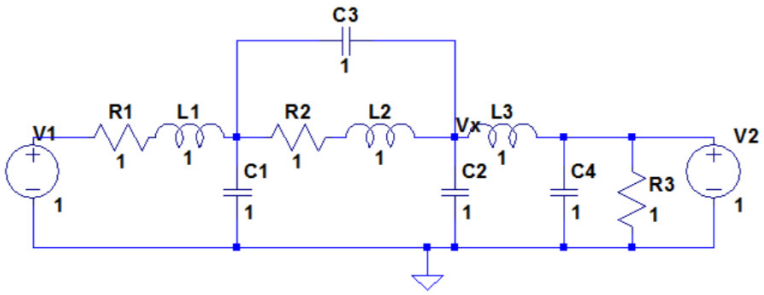
$$= \begin{bmatrix} L_{11}^2 & & \\ L_{21} \cdot L_{11} & L_{21}^2 + L_{32}^2 & \\ L_{31} \cdot L_{11} & L_{31} \cdot L_{21} + L_{32} \cdot L_{22} & L_{31}^2 + L_{32}^2 + L_{33}^2 \end{bmatrix}$$

j, Symmetric

$$a_{jj} = L_{jj}^2 + \sum_{k=1}^{j-1} L_{jk}^2$$

$$a_{ij} = \sum_{k=1}^{j-1} L_{ik} L_{jk} + L_{ij} L_{jj}$$

1. Let (G+sC)V=I.



(a) (15%) Derive the MNA equations for the above circuit.

Perform transient simulation at **Vx** by using different methods below. Let time step 0.1 second and simulate for 20 seconds. Plot the results in one figure.

(b) (15%) Use **eigenvector method**

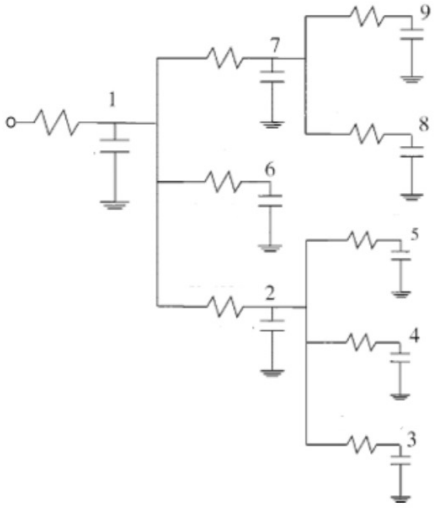
(c) (15%) Use **trapezoidal method**.

(d) (15%) Use **SPICE**.

(a) (10%) Use **backward Euler method**.

(b) (10%) Use **SPICE**.

3. (20%) Given a RC tree with several wire segments, where Wi is the width and Li is the length of each wire. The value of each Wi is between 1 and 4, and all Li are 1. The delay of each segment is Ri*Ci where resistance Ri=Li/(Wi*Thickness) and capacitance Ci=Wi*Li. Use Matlab optimization toolbox to minimize the delay at node 4. Please show the **width of each segment** and **the value of the best delay**.



*Please compress the files to a rar or zip file and e-mail to

$R = \rho \frac{l}{A}$

4. (20%) Wire sizing with nonlinear programming

We use L shape (i.e. 1 resistor and 1 capacitance) to model a wire segment in this proplem. Now given a network as shown in **Fig.4**. There are 9 wire segments, where Wi is the width, Li is the length and Hi is height of each wire (i is from 1 to 9). The resistance of each segment is Li/(Wi×Hi). The capacitance of each segment is Wi×Li. Use Elmore delay model and decide the width of each segment to minimize the delay.

(a) (6%) Please write down the delay of node 1, node 2 and node 3 using Elmore delay model. Note that you should use only Wi, Li and Hi to represent your answer (i is from 1 to 9).

(b) (14%) All Li and Hi are 1. The value of each Wi is between 1 and 10. Please find out Wi of each wire segment to minimize the total delay of all leaf nodes (i.e. node 3,4,5,6,8 and 9) with Matlab. Please show Wi of each segment and the value of the best delay. Also write down the answer on the answer sheet. (M4.m)

***Files to commit: M4.m**

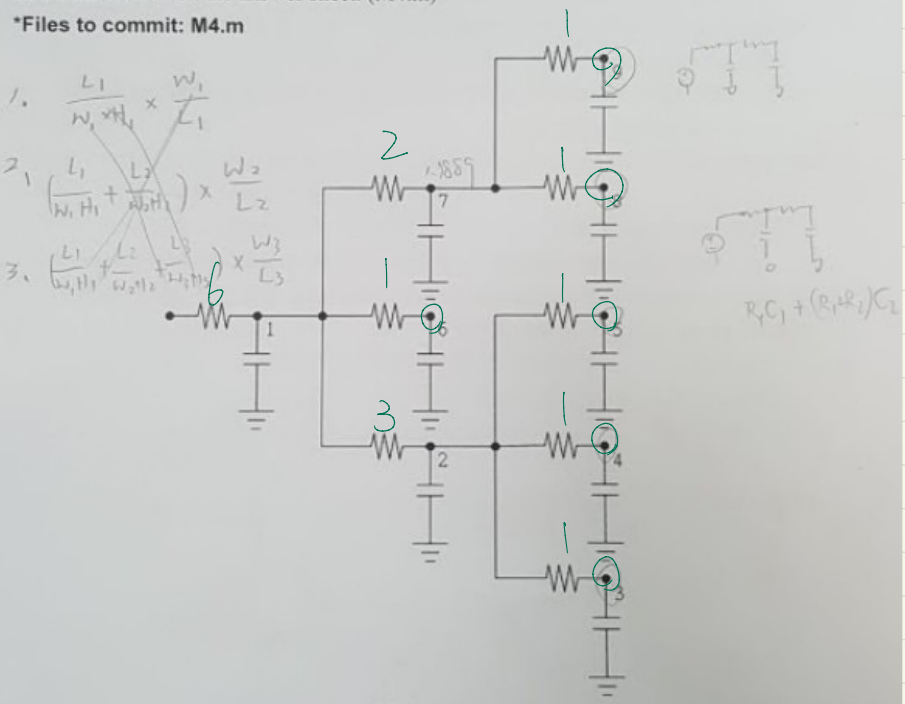


Fig.4

R08

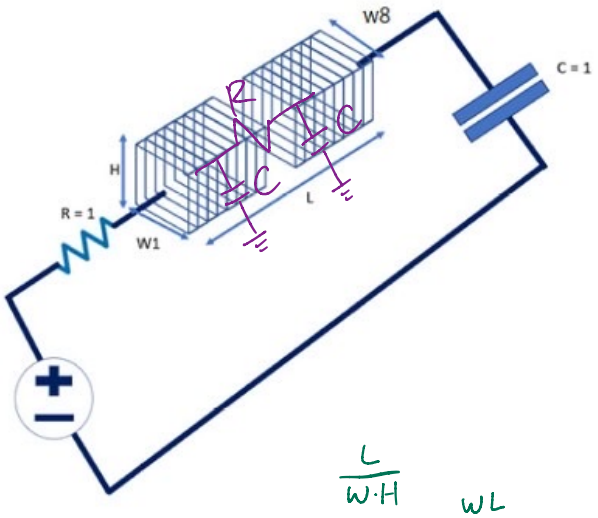
$\rho \frac{l}{A} \quad \frac{L}{w \cdot H}$

$\varepsilon \frac{A}{d} \quad w \cdot L$

(a) node 1: $\frac{L_1}{W_1 \cdot H_1}\left(\sum_{i=1}^{9} W_i L_i\right)$

node 2: $\frac{L_1}{W_1 \cdot H_1}\left(\sum_{i=1}^{9} W_i L_i\right) + \frac{L_2}{W_2 \cdot H_2}\left(\sum_{i=2}^{5} W_i L_i\right)$

node 3: $\frac{L_1}{W_1 \cdot H_1}\left(\sum_{i=1}^{9} W_i L_i\right) + \frac{L_2}{W_2 \cdot H_2}\left(\sum_{i=2}^{5} W_i L_i\right) + \frac{L_3}{W_3 \cdot H_3}(W_3 L_3)$

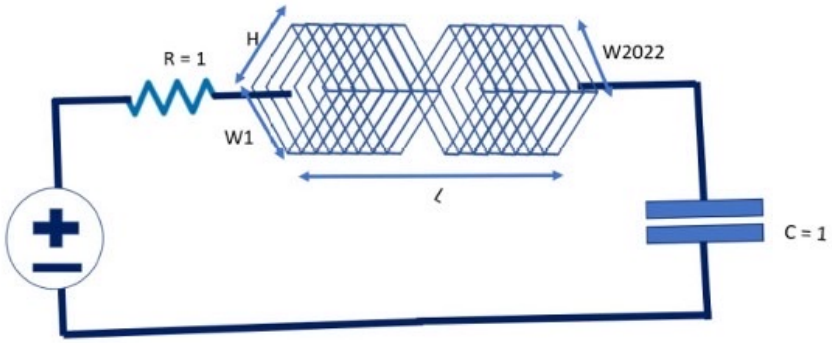


$\frac{L}{W \cdot H}$ WL

(a) **Derive the Delay equation for above circuit.** $R \propto L/W$, $C \propto$ ~~L/W~~

(b) **Find the minimum delay.**

Note: 1 ≤ Wi ≤ 8 (i form 1 to 8), H= 1, L= 8



(c) **Find the minimum delay for above circuit.**

Note: 1 ≤ Wi ≤ 8 (i form 1 to 2022) , H= 1, L= 2022

$$Delay = \left(\sum_{i=1}^{8} W_i + 1\right) + \sum_{i=1}^{8} \left(\frac{1}{W_i}\left(\frac{W_i}{2} + \sum_{j=1}^{8} W_{j+1}\right)\right)$$

AWE：用展開点附近的moments直接组pade'近似來降階，高階時容易數值不穩，且不保證被動/stable

prima：用Krylov/Arnoldi投影（$G_r = V^T G V$, $C_r = V^T C V$）做降階，通常可保留RLC網路的被動性，因此更穩、較不會發散

GE 高斯消去法

1. LU、Cholesky
2. MNA / NA (L、C)x2
3. Forward / Backward Euler
4. Eigenvector methods
5. prima vs. AWE (moment matching) 比較

稀疏矩陣

6. wire sizing
7. CG、PCG

progam

100 x 100 node matlab GCB